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Evolving cybersecurity frontiers: A comprehensive survey on concept drift and feature dynamics aware machine and deep learning in intrusion detection systems

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ABSTRACT

Intrusion Detection Systems (IDS) have become pivotal in safeguarding information systems against evolving threats. Concurrently, Concept Drift presents a significant challenge in machine learning, affecting the adaptability and accuracy of predictive models in dynamic environments. Understanding the synergy between IDS and Concept Drift is crucial for developing robust security systems. The motivation behind this survey is driven by the emerging complexities in cyber threats and the dynamic nature of data streams, which necessitate advanced IDS capable of adapting to Concept and Feature Drift. Our analysis reveals a glaring omission in the existing literature—the integration of Concept Drift and Feature Drift within IDS. Most studies have focused on Concept Drift in a general context or on IDS but have yet to comprehensively consider the implications of data dynamics. This oversight has led to a fragmented understanding and suboptimal approaches to tackling modern cyber threats. To address this, we propose a comprehensive review that delves into the role of machine learning in IDS, explicitly focusing on Concept and Feature Drift. We have proposed a framework that includes all the necessary components for a drift-aware IDS. The framework incorporates dynamic feature selection, adaptive learning algorithms, and continuous monitoring techniques to handle Concept Drift and Feature Drift effectively. The survey highlights state-of-the-art methodologies and current challenges in integrating these concepts. The methodology involves an exhaustive analysis of published works from 2019 to 2024, comparing and contrasting various models and approaches. This includes a detailed examination of Concept Drift-aware IDS methods, dynamic feature selection techniques, and the impact of high dimensionality in IDS. These quantitative improvements underscore the necessity for developing adaptive and resilient IDS. The survey uncovers under-represented areas in current research, paving the way for future investigations. By highlighting these gaps and providing comparative data, the survey sets a clear direction for upcoming research efforts to foster the development of more dynamic and adaptable IDS solutions. The quantitative experimental evaluation of the proposed framework is planned to be conducted in a future article, where we will assess its effectiveness and performance in real-world scenarios.

1. Introduction

In recent years, the frequency and sophistication of cyber threats and data breaches have alarmingly increased, impacting organizations in various sectors (Ashraf et al., 2022). These incidents often result in devastating consequences, from financial losses to reputational damage and national security risks. In this context, Intrusion Detection Systems (IDS) have become fundamental in the cybersecurity infrastructure of modern enterprises (Musleh et al., 2023). While somewhat effective in detecting known threats, traditional rule-based IDS models frequently

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fall short in adaptability and scalability. This makes them less effective against novel and evolving threats (Bhavsar et al., 2023). Moreover, they are often prone to generating high numbers of false positives and negatives, shifting focus away from real threats (Aldallal, 2022).

The emergence of Machine Learning (ML) has transformed various scientific and technological fields, and its integration into IDS shows excellent promise (Talukder et al., 2023). ML algorithms, designed to learn from data, enable more nuanced and adaptive threat detection. This learning ability allows ML-based IDS to reduce false positives and adapt more efficiently to new threats (Kaushik et al., 2023). However, implementing ML algorithms in IDS has challenges, notably Feature Drift and Concept Drift. These occur when the statistical properties of the data evolve, necessitating continual adaptation of the ML models to maintain effectiveness (Bayram et al., 2022). For IDS, this implies that a model trained on historical data might only perform well on new, unseen data if it includes mechanisms for model updating (Shyaa et al., 2023).

Understanding and addressing Concept Drift and Feature Drift within IDS is crucial for developing robust and adaptive security systems capable of responding to the dynamic nature of cyber threats. Further advancements in the field have been made by researchers addressing various aspects of IDS optimization and resilience. The study (Tao et al., 2024) investigates iterative learning control (ILC) for systems with communication constraints, which is relevant to optimizing IDS communication efficiency in resource-constrained environments (Song et al., 2023). addresses state estimation under Denial-of-Service (DoS) attacks, proposing strategies to enhance IDS resilience and reliability by managing intermittent attacks while reducing network load (Zhang et al., 2023). explores fuzzy filtering for nonlinear systems under dual cyber-attacks, offering approaches that could be adapted for IDS to improve detection accuracy under complex attack scenarios. The study (Author Anonymous et al., 2024 examines heterogeneous networks (HetIoT). It proposes the ONIDS online NID technique, which includes a Beta distribution-based inference technique and an online ID technique called ELMO. These methods address class imbalance and concept drift in IoT environments, making the IDS more effective for online and offline NID applications. Another significant contribution is by (Li et al., 2024, who adopted the ATT&CK matrix for mapping unstructured threat intelligence to tactics and techniques. Using DistilBERT-based classifiers, this method improves the accuracy of threat detection and classification, demonstrating the effectiveness of leveraging advanced ML techniques in IDS. These studies collectively highlight the ongoing efforts to enhance IDS through innovative methodologies that address both traditional challenges and emerging threats, emphasizing the critical role of continuous adaptation and improvement in cybersecurity systems.

The article's organization systematically explores the intersection of Machine Learning (ML) and Intrusion Detection Systems (IDS), specifically focusing on addressing the challenges posed by feature and concept drift. Initially, a foundational understanding of IDS is provided, where the evolution and integration of ML in IDS are detailed, especially emphasizing stream data mining and abstraction learning models in data streams. This is followed by a comprehensive literature review that critically analyzes Concept Drift-Aware IDS Methods, dynamically explores feature selection strategies, and evaluates the role of reinforcement learning in these processes. Subsequently, a novel framework is introduced, specifically designed to mitigate the challenges identified in IDS, including concept drift and feature drift, The article then discusses the proposed framework in detail, introducing the experimental design, existing datasets, and evaluation metrics. Recommendations for enhancing threat detection systems are provided, highlighting the need for continuous monitoring techniques and the implementation of adaptive learning algorithms. The article then outlines the primary research challenges and limitations in the current IDS landscape, focusing on the joint handling of concept drift and feature drift, the complexities of managing concept drift alongside data imbalance, and the difficulties in detecting intricate drifts. The article concludes with a summation of the critical insights, the contributions of the proposed framework, and prospective directions for future research in the domain of IDS and ML, particularly concerning adaptation and mitigation strategies for feature and concept drift.

1.1. Relation between IDS and Concept Drift

Intrusion Detection Systems (IDS) are critical in cybersecurity and are tasked with identifying unauthorized or abnormal activities within networks or systems. These systems strive to accurately detect attacks, malware, or unauthorized data exfiltration (Halbouni et al., 2022b). As the cybersecurity threat landscape evolves, the concept of 'Concept Drift' becomes increasingly relevant for IDS (Andresini et al., 2021). Concept Drift is the phenomenon where the statistical properties of the target variable, the observed features, or the decision boundaries change over time (Agrahari and Singh, 2022b). This is particularly pertinent in machine learning, where models are trained on historical data to predict or decide based on new data. If the underlying concept shifts over time, the model's effectiveness in making accurate predictions can be compromised (Suárez-Cetrulo et al., 2023). In IDS, concept drift manifests in various forms, each posing unique challenges. Signature-based Drift, for instance, includes issues like Protocol Mutation, where obsolete protocols are replaced by new ones (Apruzzese et al., 2023), and Signature Variations, where attackers modify known patterns to avoid detection (Kuppa and N. A. Le-Khac, 2022). Anomaly-Based Drift involves changes in User Behavior, such as the shift to remote work altering network traffic patterns (Kermenov et al., 2023; Soltani et al., 2024), and Infrastructure Changes that redefine 'normal' network behavior (Thakkar and Lohiya, 2023). Hybrid IDS models, combining anomaly-based and signature-based methods, confront the added complexity of Mixed Strategies, entailing multiple layers of potential concept drift (Abdulganiyu et al., 2023a). In the context of Application-Specific Drift, the increased use of IoT Devices introduces new network interactions not covered by older IDS models (Yang and Zhang, 2023). Geopolitical Changes, like the evolution of State-Sponsored Attacks (Khraisat et al., 2019), drive concept drift influenced by geopolitical dynamics. Lastly, Regulatory Changes, such as adjustments in Data Privacy Laws, necessitate changes in data collection and processing, leading to shifts in the metrics and features for intrusion detection (Hewage et al., 2023). We present the various types in Table 1. In the next section, we present the research gap and the contribution.

1.2. Research gap and contribution

Table 2 presents an overview of the existing surveys from the perspective of reviewed models, IDS, and concept drift focus. A critical gap in existing literature has been identified, focusing on the challenges of Concept Drift (CD) and Intrusion Detection Systems (IDS). While several surveys have provided comprehensive insights into various facets of either CD or IDS, a unified approach addressing both challenges has been notably absent. Published works from 2020 to 2023 have been examined, revealing a predominant focus on machine learning and deep learning methodologies for tackling Concept Drift. However, integrating these dynamic models within IDS needs to be more noticed. Moreover, these studies need to pay more attention to the Feature Drift (FD) issue within the scope of IDS. Thus, there is a pressing need for a comprehensive survey. Synthesizes knowledge across these domains and has been recognized, setting the stage for future research endeavours.

This article's contribution is twofold: it addresses a notable gap in current literature by providing a holistic analysis of IDS in the context of Concept Drift. It proposes a framework for future research to develop more resilient and adaptive IDS. We state the details of the contributions as follows.

Table 1
An overview of various types of Drift and their relation to IDS.

Type of Drift	Example	Description
Signature-Based Drift	Protocol Mutation (Apruzzese et al., 2023)	Older protocols get phased out, causing static IDS models to fail in recognizing new or updated protocols.
	Signature Variations (Kuppa and N. A. Le-Khac, 2022)	Attackers modify known attack signatures to evade detection, causing effectiveness to decrease over time.
Anomaly-Based Drift	User Behavior (Kermenov et al., 2023; Soltani et al., 2024)	Significant changes in typical user behavior, such as remote work, may alter network traffic patterns.
	Infrastructure Changes (Thakkar and Lohiya, 2023),	Upgrades or changes in network infrastructure alter what is considered 'normal', affecting IDS models.
Hybrid Models	Mixed Strategies (Abdulganiyu et al., 2023b)	IDS employing both anomaly-based and signature-based methods face multiple layers of concept drift.
Application-Specific	IoT Devices (Yang and Zhang, 2023)	The increasing use of IoT devices changes network interactions, causing older IDS models to become obsolete.
Geopolitical Changes	State-Sponsored Attacks (Khraisat et al., 2019),	New forms of Advanced Persistent Threats (APTs) introduce concept drift due to geopolitical shifts.
Regulatory Changes	Data Privacy Laws (Hewage et al., 2023)	Changes in data privacy laws can alter data collection methods, causing a Drift in metrics used for IDS.

1. Employing a systematic review methodology, our survey presents an exhaustive examination of the integration of machine learning in IDS, with a particular emphasis on the implications of Concept and Feature Drift. This includes a detailed discussion on stream data mining, abstraction learning models, and the role of various concept drift detectors. A critical review of methodologies, such as ensemble learning, dimensionality reduction through Principal Component Analysis (PCA), Fisher Discriminant Analysis, and Information Gain, provides a nuanced understanding of the current state-of-the-art.
2. By dissecting the current literature and synthesizing knowledge across the domains of IDS and CD, we identify the absence of a unified model that addresses IDS adaptability and Concept Drift's dynamism. Furthermore, we highlight the under-researched area of Feature Drift within IDS. This element is crucial for maintaining the efficacy of detection systems in the face of changing data patterns.
3. 3-Through identifying these gaps, the survey delineates a clear research trajectory for future studies. It emphasizes the need for innovative approaches that combine the detection of complex drift types with the robustness of IDS. The article also points out the scarcity of literature on the joint handling of concept drift and data imbalance and the detection of complex types of drifts, thus encouraging researchers to explore these areas.
4. To advance the field, we propose a conceptual framework for IDS that can adapt to Concept and Feature Drift. The framework suggests a multi-layered approach incorporating dynamic feature selection, reinforced by reinforcement learning, to enable IDS to self-adjust in real time. The framework also considers the challenges of high dimensionality, advocating for sophisticated feature extraction and selection techniques to improve detection accuracy in data-rich environments.

The section concludes by asserting the necessity for an integrated approach to IDS design that is cognizant of and responsive to Concept Drift's implications. Thus, our contribution fills a significant gap in the literature and serves as a cornerstone for the next generation of IDS equipped to handle the complexities of modern cybersecurity challenges.

1.3. *Structure and abbreviations*

The structure of this survey is delineated as follows, with a visual representation provided in Fig. 1. Section 1 introduces the relationship between Intrusion Detection Systems (IDS) and concept drift, identifies research gaps, and outlines the survey's contributions. Section 2 offers a

Table 2
An overview of the existing surveys from the perspective of models discussed IDS and Concept Drift focus.

Related Surveys	Year	Concept Drift	Feature Drift	High Dimensionality	IDS, Cyber-Security	Data stream
(Bayram et al., 2022)	2022	√	×	×	×	√
Agrahari and Singh (2022b)	2022	√	×	×	×	√
Xiang et al. (2023)	2023	√	×	×	×	√
Han et al. (2022)	2022	√	×	×	×	√
(Momand et al., 2023)	2023	×	×	×	√	×
(Suárez-Cetrulo et al., 2023)	2023	√	×	×	×	√
(Abdulganiyu et al, 2023a)	2023	×	×	√	√	×
Alkasassbeh and Al-Haj Baddar (2023)	2023	×	×	×	√	×
Yi et al. (2023)	2023	×	×	×	✓	×
Kheddar et al. (2023)	2023	×	×	×	✓	×
Adnan et al. (2021)	2021	✓	×	✓	✓	√
Ahmad et al. (2021)	2021	×	×	×	√	×
Martins et al. (2022)	2022	✓	×	×	✓	√
Halbouni et al. (2022b)	2022	×	×	×	√	×
Thakkar and Lohiya (2023)	2023	✓	×	×	√	×
Wang et al. (2023)	2023	✓	×	×	√	×
Lima et al. (2022)	2022	✓	×	×	×	√
Our work		✓	✓	✓	✓	√

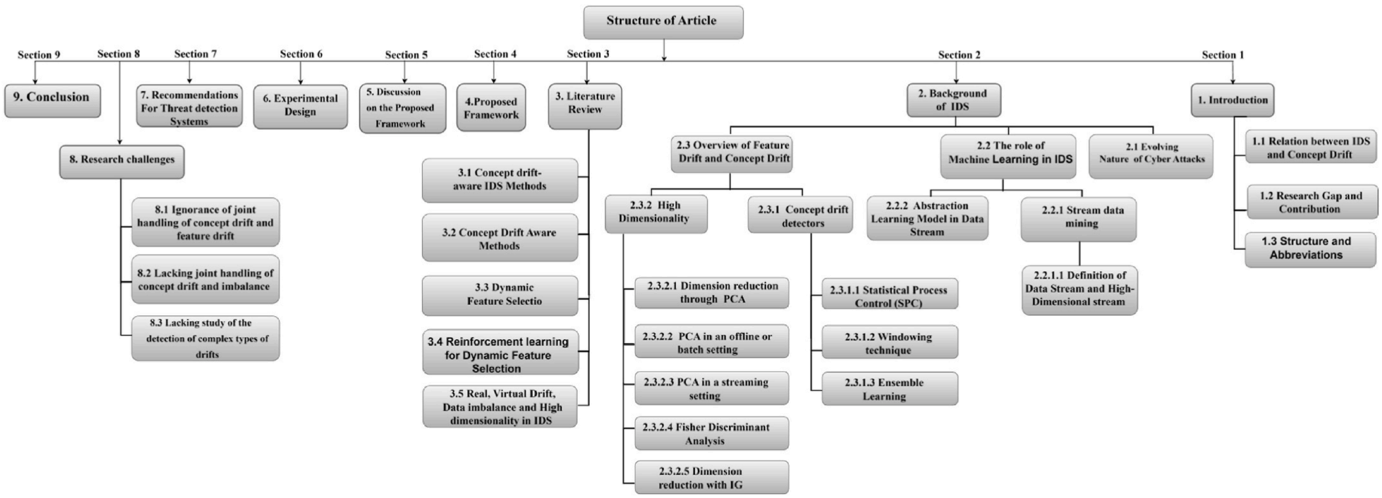

Fig. 1. The organization and structure of the survey.

background overview of IDS, discussing the evolving nature of cyber-attacks and the role of machine learning in IDS, including abstraction learning models and stream data mining. Section 3 comprises a literature review focusing on the limitations of current works, covering feature drift, concept drift, high dimensionality, and various concept drift detectors. The proposed framework for drift-aware IDS is detailed in Section 4, which includes methods for dynamic feature selection and reinforcement learning. Section 5 offers a discussion of the proposed framework. Section 6 introduces the experimental design, existing datasets in IDS, and evaluation metrics. Section 7 provides recommendations for threat detection systems. Section 8 introduces new ideas and directions for addressing challenges associated with IDS, such as joint handling of concept and feature drift. The concluding remarks are found in Section 9. Additionally, Table 3 summarizes the acronyms used throughout the survey.

2. Background of IDS

Intrusion Detection Systems (IDS) have played an integral role in cybersecurity for the last two decades. The 2000s marked a significant shift in the intrusion detection approach, with machine learning techniques becoming increasingly popular. Early IDS, during the late 1990s and early 2000s, primarily relied on signature-based techniques, which used predefined rules or patterns to identify known attacks. However, these traditional IDS were less effective at detecting novel or previously unseen attacks (Khraisat et al., 2019).

IDS functions based on either signature-based or rule-based techniques when detecting and analyzing attacks. In signature-based IDS, the system matches network data with the attack patterns stored in its database. An alarm is generated (Kheddar et al., 2023). One major drawback of signature-based Intrusion Detection Systems (IDS) is that they can only detect attacks that are present in the IDS database. On the other hand, anomaly-based IDS, also known as rule-based systems, store normal behavior states in the IDS database. These systems constantly monitor all the events happening on the network and raise the alarm whenever a deviation is outside the specified rules. Anomaly-based IDS have a significant advantage over signature-based IDS as they can detect unknown attacks (Kilincer et al., 2021). In Fig. 2, the taxonomy of IDS systems is provided.

However, these systems suffered from high false-positive rates, leading to the need for more sophisticated methods. Anomaly-based Intrusion Detection Systems (IDS) define a ’normal’ behaviour profile based on historical data and then consider deviations from this ’normal’ as potential intrusions. This approach can effectively identify previously unknown attacks, a significant advantage over signature-based systems that only detect known threats (Alkasassbeh and Al-Haj Baddar, 2022).

However, the challenge with anomaly-based systems arises from the difficulty in accurately defining what ’normal’ behaviour is, as it can vary significantly across different systems and over time. Additionally, ’normal’ behavior in a network is not always wholly benign, and malicious activities can sometimes mimic ’normal’ behavior patterns, further complicating the detection process. Moreover, many legitimate activities can appear anomalous in specific contexts. For instance, a system administrator performing maintenance tasks may perform actions deviating from their typical behavior, triggering false alarms. Likewise, changes in network traffic patterns due to non-malicious reasons, such as increased load or software updates, can also be mistakenly flagged as anomalies (Albasheer et al., 2022).

All these factors contribute to a high rate of false positives -benign activities incorrectly flagged as malicious-in anomaly-based IDS. High false-positive rates can be problematic as they lead to alarm fatigue, where security personnel are overwhelmed with alerts, potentially causing them to overlook real threats. This is why there is a continuous effort in the field to develop more sophisticated methods that maintain the strengths of anomaly-based IDS while reducing the rate of false positives. These efforts often involve using advanced machine learning and statistical techniques to improve the accuracy of ’normal’ behavior modeling and the identification of genuine anomalies (Dini et al., 2023).

In the last decade, the application of machine learning techniques in IDS has grown, focusing on reducing false-positive rates and improving detection accuracy. Researchers have explored various machine learning models, such as decision trees (Le et al., 2022), support vector machines (Ahmad et al., 2022), and neural networks (Halbouni et al., 2022a), among others, for intrusion detection. The most recent advancements in IDS include the application of deep learning (Simon et al., 2022)and reinforcement learning models (Tharewal et al., 2022), aiming at further improving the detection capabilities and handling the increasing complexity of network data (Momand et al., 2023.

2.1. Evolving nature of cyber attacks: understanding attack development life cycle

The attack development life cycle is a crucial method for describing the process of conducting cyber-attacks. Understanding how attackers operate is essential for developing effective defensive strategies. Various cyber-attack frameworks are used to represent adversarial actions and behaviors, each providing insights into different stages and techniques of cyber threats. The most well-established models for cyber threat representation include.

Table 3
Acronyms are used in the article.

CD	Concept Drift	FTRL-ADP	Follow-the-Regularized-Leader with Adaptive Decaying Proximal
FD	Feature drift	NCM	Nearest Class Mean
ML	Machine learning	HDDM	Hoeffding Drift Detection Method
IRL	Interactive reinforcement learning	HDWM	Heterogeneous Dynamic Weighted Majority
DRLFS	Deep Reinforcement Learning-based Feature Selector	HLFR	Hierarchical Linear Four Rates
DFBFS	Dynamic Filter-Based Feature Selection	KME	Knowledge-maximized Ensemble
ABFS	Adaptive Boosting for Feature Selection	MLP-GA	Multilayer perceptron with genetic algorithm
IG	Information Gain	LFR	Linear Four Rates
FA-OSELM	Feature adaptive -Online sequential extreme learning machine	MD3	Margin Density Drift Detection
ACCD	Associative Classification over Concept Drifting Data Streams	MDDM	McDiarmid Drift Detection Method
ACDDM	Accurate Concept Drift Detection Method	meta-RRKOS-	Meta-cognitive Recurrent Recursive Kernel
OAUE	Online Accuracy Updated Ensemble	SAW_WDA	Secure adaptive windowing and website data authentication
ALD	Approximate Linear Dependence	ELM-DDM	Kernel Online Sequential Extreme Learning
AUE	Accuracy Updated Ensemble	MOS-ELM	Meta-cognitive online sequential extreme learning machine
PWPAE	Performance Weighted Probability Averaging Ensemble	ADDM	Adaptive sliding window-based Detection Method
AWE	Accuracy Weighted Ensemble	ADDS	Anti-concept Drift Detection Algorithm
CDTs	Change Detection Tests	ADMIN	Adaptive Windowing
CSDD	Cosine Similarity Drift Detector	ADTCD	Adaptive anomaly detection model toward concept drift
CUSTOM	Cumulative SUM	NB	Naive Bayes
DDD	Diversity for Dealing with Drifts	NDE	Number and Distance of Errors
DDM	Drift Detection Method	NSE	Nonstationary environments
DDM-OCI	Drift Detection Method for Online Class Imbalance	FDA	Fisher's Discriminant Analysis
KP-OSELM	Knowledge Preserving-Online sequential extreme learning machine	OMR-DDM	Online Map-Reduce Drift Detection Method
DELM	Dynamic extreme learning machine	OS-ELM	Online sequential extreme learning machine
DOED	Diversified Online Ensembles Detection	OWE	Online Weighted Ensemble
DWM	Dynamic Weighted Majority	PHT	Page-Hinkley test
ECDD	EWMA for Concept Drift Detection	AGE	Accuracy and Growth rate updated Ensemble.
ECHO	Efficient Concept Drift and Concept Evolution Handling over Stream Data	LSTM	Long Short-Term Memory
DUE	Dynamic Updated Ensemble	PINE	Predictive and Parameter Insensitive Ensemble
EGBDT	Elastic gradient boosting decision tree	PSO	Particle Swarm Optimization
EnDDM	Ensemble Drift Detection Method	RDDM	Reactive Drift Detection Method
ELM	Extreme learning machine	DNN	Deep neural network
ESOS-ELM	Ensemble of online sequential extreme learning machine	RDWM	Recurring Dynamic Weighted Majority
EWAUC	Equal Weighted AUC	SEA	Streaming Ensemble Algorithm
MAFSIDS	Multi-Agent Feature Selection Intrusion Detection System	SPC	Statistical Process Control
EWMA	Exponentially Weighted Moving Average	STEPD	Statistical tests of equal proportions
FDR	False Discovery	SVM	Support Vector Machines
CNN	Convolutional neural network	MLPGDT	Multilayer perceptron gradient decision tree
FHDDM	Fast Hoeffding Drift Detection Method	WELM	Weighted extreme learning machine
FHDDMS	Stacking Fast Hoeffding Drift Detection Method	WMA	Weighted Majority Algorithm
FsNB	Fast switch Narve Bayes model	AdIter	Adaptive Iterations
PCA	Principal Component Analysis	WSTD	Wilcoxon Rank Sum Test Drift Detector
FTRL	Follow the Regularized Leader	GPC	Genetic programming combiner
EDIT	Error Distance for drift detection and monitoring	ECPF	Enhanced Concept Profiling Framework
DQFSA	Deep Q-learning-based Feature Selection Architecture	EDDM	Early Drift Detection Method
DRFSA		Dynamic recursive feature selection algorithm	

1. **Attack Trees:** Introduced in 1999 by Bruce Schneier, Attack Trees model security threats by illustrating all the possible ways a target can be attacked. This method helps visualize an attacker's different paths, emphasizing the evolving strategies and techniques used in different scenarios (Hutchison, 2005).
2. **Cyber Kill Chain:** Developed in 2011 by Lockheed Martin analysts, the Cyber Kill Chain framework focuses on detecting adversarial intrusions across the entire cyber-attack life cycle. It breaks down the attack process into distinct stages, from reconnaissance to exfiltration, highlighting the evolving nature of threats and the need for continuous monitoring and adaptation (Hutchins et al., 2011).
3. **MITRE's Adversarial Tactics, Techniques, and Common Knowledge (ATT&CK):** MITRE's ATT&CK framework comprehensively covers pre- and post-compromise techniques. It categorizes various tactics and techniques attackers use, offering a detailed understanding of their evolving behaviors and enabling defenders to anticipate and counter these actions effectively (Strom et al., 2018).
4. **Mandiant's Attack Life Cycle Model:** This model emphasizes the typical patterns of Advanced Persistent Threat (APT) intrusions, showcasing the iterative and escalating nature of attackers' strategies to gain and maintain access to target systems. It highlights how attackers continuously refine their methods to bypass defenses and escalate privileges over time(Shavazipour et al., 2021).

These models collectively offer a knowledge base of attackers' processes, revealing cyber threats' sophisticated and evolving nature. The Cyber Kill Chain has been widely adopted as a core foundation for most existing adversary-focused models due to its structured approach to mapping the stages of an attack.

Understanding these frameworks is vital for cybersecurity professionals as they provide a structured way to analyze and anticipate adversarial actions. By studying these models, defenders can better understand the evolving tactics, techniques, and procedures (TTPs) used by attackers, allowing them to develop more robust and adaptive defence mechanisms. This comprehensive approach to understanding the attack development life cycle ensures that cybersecurity measures can evolve with the increasing sophistication of cyber threats, ultimately enhancing the resilience of organizational defences against a wide range of cyber attacks.

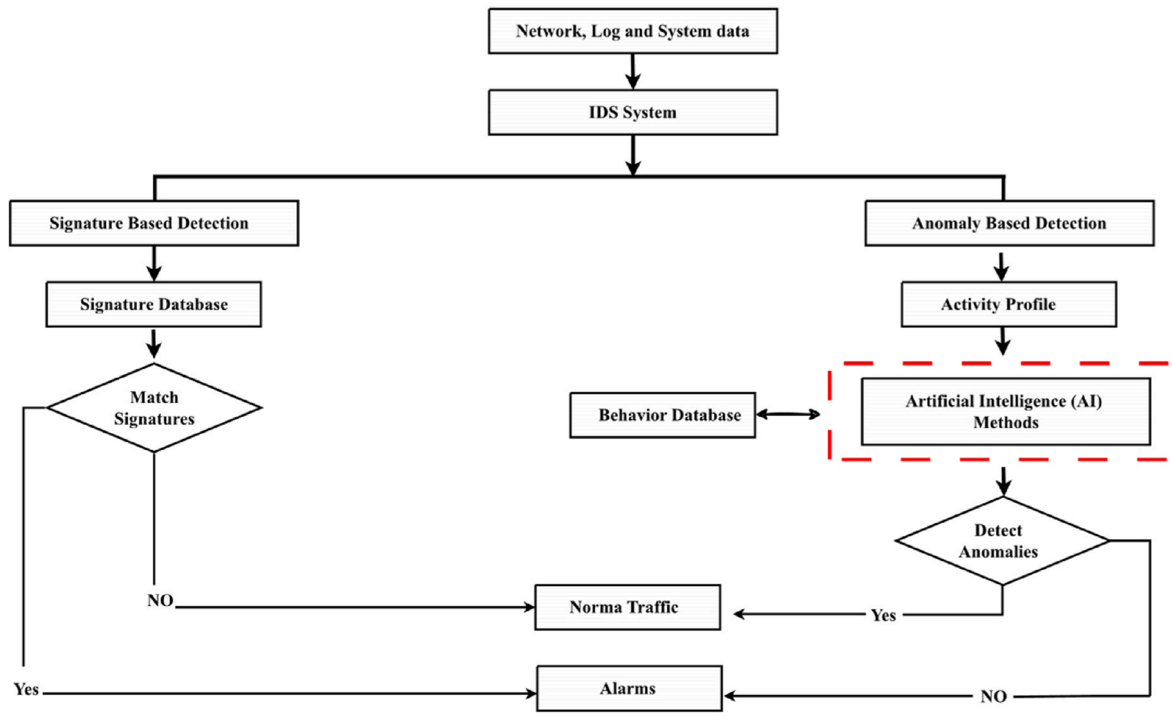

Fig. 2. Taxonomy of traditional IDS systems (Kilincer, Ertam and Sengur, 2021).

2.2. *The role of machine learning in IDS*

Artificial Intelligence (AI) has become integral to Intrusion Detection Systems (IDS), significantly enhancing their effectiveness and efficiency. As mentioned, IDSs are software applications or hardware devices that monitor a network or system for malicious activity or policy violations. They have traditionally been rule-based, relying on known intrusion signatures. However, the rapid evolution of cyber threats necessitated an improvement in these systems, a role AI has excellently filled (Dina and Manivannan, 2021).

AI facilitates anomaly detection by analyzing massive network traffic and identifying abnormal patterns that suggest potential intrusion. Using statistical analysis, clustering, and classification methods, AI and machine learning (ML) have proven substantially effective at detecting novel, previously undocumented threats. Additionally, these technologies can predict future threats based on historical data, thus allowing organizations to address potential intrusions proactively and mitigate the impact of cyberattacks (Halbouni et al., 2022b).

AI also automates threat responses, considerably reducing response times by automatically blocking IPs, isolating systems, or initiating scripts to counteract the intrusion. A key challenge in IDS is the high rate of false positives, which can lead to alert fatigue and consume considerable investigation resources. AI can dramatically improve intrusion detection accuracy, reducing false positives (Liu and Lang, 2019).

AI algorithms can also quickly process and analyze enormous volumes of data, making AI-based IDS more scalable and capable of handling the significant data traffic typical in contemporary networks. Moreover, AI learns and adapts to the constantly evolving nature of cyber threats, recognizing new malicious behavio(Apruzzese et al., 2023). It can also present insightful reports and visualizations of network traffic and potential threats, simplifying the understanding of network activities for security personnel. However, the effectiveness of AI and ML models depends on the quality of training data; hence, they need continuous updating with recent threat data. Also, the increasing complexity of AI models necessitates a higher level of management expertise (Magán-Carrión et al., 2020). This survey focuses on stream data concepts in ML due to their relationship with IDS. Hence, we present two overviews in the subsequent two subsections: stream data mining and feature and concept Drift, providing their relations to IDS.

2.2.1. *Stream data mining*

This section defines the main concepts of data streams and high-dimensional stream data and provides an abstraction of learning models in the data stream.

2.2.1.1. Definition of data stream and high dimensional stream. A data stream can be defined as a chronologically ordered series of data points, each associated with a specific time marker. The nature of this sequence is inherently indefinite, indicating an unspecified quantity of subsequent data. Characterized by their considerable size and diverse velocity, these streams are a hallmark of contemporary technology scenarios. In these contexts, many applications generate copious amounts of data at accelerated rates, demanding an analysis approach that is effectively real-time (Aguiar et al., 2023).

A data stream is characterized as an extensive, ongoing, limitless, and non-static data sequence characterized by its rapid arrival rate and a distribution that frequently varies with time. The elements within these streams can range from straightforward attribute-value pairs, akin to those in relational database tuples, to more intricate configurations, such as graphs. Instances of such data encompass web searches and sensor data, among others (Bahri et al., 2021). Formally, this can be represented as shown in Equation (1):

$$D = \{\langle X_1, t_1\rangle, \langle X_2, t_2\rangle, \ldots, \langle X_\infty, t_\infty\rangle\} \quad (1)$$

Here, D represents an unending sequence of stream tuples. Every data element $X_n = (x_n^1, \ldots, x_n^d)$ is a vector of d dimensions, with n denoting the count of data points arriving at time $t.F = \langle F_1, F_2, \ldots, F_d\rangle$ denotes the set of dimensions (attributes or features), and t_j is the timestamp that marks the present moment. Data streams introduce novel complexities for machine learning and data mining domains. This is because conventional techniques are tailored for static datasets typically housed in bounded, enduring data repositories and characterized by a stable distribution, unlike data streams. As a result, standard algorithms often fall short of meeting the unique processing demands of data streams; the most significant requirements are the following (Souiden et al., 2022:

1. **The Arrival Speed and Transition Nature**: Due to their arrival speed and transition nature, the data stream items require a one-time scan of each particular within a limited timeframe.
2. **Memory Constraints**: The infinite nature of the data stream means that storing and computing all the data is impossible, so only a small portion can be retained.
3. **Curse of Dimensionality**: The data are subject to the curse of dimensionality, which requires incorporating methods for handling it. Techniques such as dimensionality reduction (e.g., Principal Component Analysis, Linear Discriminant Analysis), feature selection, and manifold learning are often employed to mitigate this challenge.
4. **Class Imbalance**: The data suffers from imbalance regarding the sample's distributions over class. Techniques like resampling (oversampling the minority class or undersampling the majority class), cost-sensitive learning, and synthetic data generation (e.g., SMOTE) are commonly used to address this issue.
5. **Distribution Changes**: The distribution of the items may change over time, so the model needs to be updated periodically as past data may become irrelevant.

Scholars in this field are actively working to modify existing algorithms to fit these unique needs or develop new, specialized ones. Various summarization techniques have been devised to comply with the constraints presented by data streams (Mirzaie et al., 2023). These include windowing methods (Verwiebe et al., 2022), sampling strategies (such as reservoir sampling) (Karras et al., 2022), and others, all designed to address the challenges of memory and time constraints. Additionally, several tactics have been developed to manage the non-stationarity of data streams. These include specific concept drift detectors and forgetting mechanisms, which discard outdated data instances, among other strategies. Windowing, in particular, is a prevalent method for data stream summarization and handling drift; it focuses on a segment of the stream and applies a recency-based forgetting mechanism (Bayram et al., 2022). Further details on these techniques will be presented in the following subsection.

In specific applications, data stream objects may be characterized by 10 or more attributes, sometimes extending to hundreds, thousands, or even millions. Data of this nature, with $d \geq 10$, is classified as high-dimensional data streams (Souiden et al., 2022). Examples of such data include credit card transactions (Zioviris et al., 2022), multimedia streams (Ahmed et al., 2023), wireless sensor network data (Xu et al., 2023; Grote-Ramm et al., 2023), and robotics (Lesort et al., 2020). In intrusion detection, the analysis requires simultaneous consideration of multiple attributes or features, such as source and destination IP addresses, port numbers, protocol types, payload content, and timestamps. This combination of diverse attributes typically results in high-dimensional data. Furthermore, Intrusion Detection Systems (IDS) need to analyze this data within a temporal framework. Given the rapid changes in network traffic, these data streams are high dimensional and highly dynamic. Continuous monitoring in this environment involves identifying patterns and anomalies over time, thus introducing an additional layer of complexity (Soltani et al., 2024).

2.2.2. Abstraction learning model in data stream

Data stream learning can be performed in three ways: supervised, semi-supervised, and unsupervised. As it is depicted in Fig. 3 the abstract view of model building in the data stream consists of three modules.

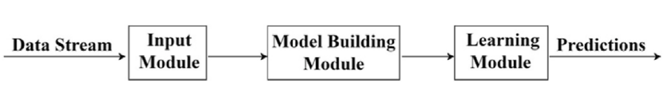

Fig. 3. An abstract view of stream data learning.

- **Input Module**: Takes input of training data instances/examples and stores them in memory for later processing.
- **Model Building Module**: Builds the initial model, taking new incoming data instances and predicting their labels.
- **Learning Module:** Learns from the current data pattern, providing the basis for model building.

Understanding the differences between traditional data mining and stream data mining is crucial. In Table 4, we compare Traditional data mining and data stream mining from various perspectives, such as Passes Required, Time Constraints, Memory Usage, Concept Diversity, and Accuracy of Results. (Agrahari and Singh, 2022b).

Traditional data mining allows for multiple passes, unlimited time, and memory, focusing on one concept and aiming for accuracy. In contrast, data stream mining operates with a single pass, limited time, and memory, considering more than one concept and often resulting in approximate outcomes. This restructuring aligns with the requested order and provides a clear and comprehensive overview of the subject matter. If further adjustments or elaborations are needed, please let me know.

2.3. Overview of feature drift and Concept Drift

This section presents an overview of the challenges that occur in IDS. More specifically, we present an overview of each type of concept drift, concept drift detectors, and high dimensionality.

Definition of Concept Drift: For a given data stream S, represented as a sequence of input attributes and class labels $(x(t), y(t))$, concept drift may occur between two points in time, t *and* $t + D$, if and only if there exists x such that $p(ix,y) \neq p_{t+D}(x,y)$, where refers to the point distribution at time t between the set of input attributes and the class label (Wares et al., 2019.

As the definition states, Concept drift refers to the phenomenon where the statistical properties of the target variable change over time. In the context of IDS, concept drift can be particularly challenging as it may lead to changes in the patterns of normal and malicious activities (Karimian and Beigy, 2023). According to the definition of concept drift and the characteristics of joint probability, concept drifts have three causes.

1. Virtual Concept Drift

In the context of Virtual Concept Drift (Feature drift), the key characteristic is the change in the probability distribution of the feature set ' x ' over different time intervals, while the conditional probability of the target variable ' y ' given ' x ' does not vary (Barddal et al., 2017). Mathematically, this scenario is represented as $P_{t_0}(x) \neq P_{t_1}(x)$ and $P_{t_0}(y|x) = P_{t_1}(y|x)$. During such occurrences of virtual concept drift, the decision boundary of the model is not impacted, even though there are changes in the feature space distribution (Suárez-Cetrulo et al., 2023; Barddal et al., 2019b).

For example, considering the feature of the source IP address in network traffic, a change in the distribution of source IP addresses without a corresponding change in the maliciousness of the traffic (i.e.,

Table 4
Comparative analysis of traditional data mining vs data stream mining.

Aspect	Traditional Data Mining	Data Stream Mining
Passes Required	Multiple	Single
Time Constraints	Not restricted	Restricted (Real-time)
Memory Usage	No limitation	Constrained
Concept Diversity	Single	Multiple
Accuracy of Results	High precision	Estimative

the label y) would be a virtual drift. This might occur because a new subnet is added to the network but does not necessarily indicate a change in the underlying threat landscape (Prasath et al., 2022).

2. Real Concept Drift

In instances of Real Concept Drift, a significant change is observed in the conditional probability of the target variable ' y ' given the feature set ' x ', while the probability distribution of ' x ' remains steady. This is mathematically denoted by $P_{t_0}(y|x) \neq P_{t_1}(y|x)$ and $P_{t_0}(x) = P_{t_1}(x)$. Such a change directly impacts the prediction model, signifying a true concept drift. This alteration affects not just the distribution within the feature space but also the decision-making boundaries of the model (Han et al., 2022).

For example, assuming that the feature represents the number of failed login attempts to a system if a new type of brute-force attack emerges that changes the relationship between failed login attempts and malicious activity, this would constitute real concept drift. The underlying threat pattern has changed, requiring an update to the IDS model (Kuppa and N.-A. Le-Khac, 2022).

3. Hybrid Concept Drift

In environments characterized by dynamic data streams, it is possible for both real and virtual concept drifts to co-occur, i.e., $P_{t_0}(x) \neq P_{t_1}(x)$, $P_{t_0}(y|x) \neq P_{t_1}(y|x)$.

This is a mixed concept drift, which is most common. It is worth noting that according to the Bayesian decision theory (Richard et al., 2007), we obtain Equation (2):

$$P(y|x) = \frac{P(y) * P(x|y)}{P(x)} \tag{2}$$

It can be seen that $P_t(y)$ and $P_t(x|y)$ also affect $P_t(y|x)$, thus indirectly causing a real concept drift. The specific manifestations of the concept drift due to different causes are shown in Fig. 4, in which (X_1 *and* X_2) represent the two-digit feature space, and y represents its category label.

Fig. 4 illustrates different instances of concept drift arising from various causes. This illustration depicts the feature space using a two-dimensional representation, denoted as (X_1 *and* X_2), while y signifies the category label associated with each data point (Xiang et al., 2023).

For example, A hybrid drift might occur when a network undergoes significant changes, such as the addition of new devices (virtual Drift) and new attack vectors (real Drift). This complex scenario requires careful monitoring and adaptation of the IDS (Folino et al., 2020). In the following sub-section, we present the types of transitions of concepts in concept drifts.

Transition Types of Concepts in Concept Drift In the ever-evolving field of data science, the concept of 'concept drift' stands out as a critical factor, especially in the context of predictive models within dynamic environments. Concept drift occurs when there is a shift in the statistical attributes of the target variable being predicted, leading to potential declines in model performance. Adapting to these shifts is crucial for maintaining model efficacy. Recognizing and understanding the various forms of concept drift is critical to effective model management. Each form presents its own set of unique challenges and implications for model adjustment. The subsequent sections delve into the diverse types of concept drift transitions, including gradual, sudden, incremental, recurrent, and blip drifts, each affecting model adaptation strategies differently (Lu et al., 2019).

1. **Gradual Drift**: Gradual Drift refers to a slow and continuous change in the underlying concept, where two concepts, H and H′, may interchange over time before stabilizing. Unlike abrupt changes, gradual Drift represents a steady evolution in the data, as shown in Fig. 5(e) (Ramírez-Gallego et al., 2017).

The Challenges.

1. Slow Evolution: Gradual Drift is characterized by a slow and continuous transformation, making it subtle and sometimes hard to detect.
2. Interchanging Concepts: H and H′ may switch back and forth before stabilizing, adding complexity to the detection process.
3. Adaptation Timing: Due to its slow nature, determining the right time to adapt from H to H′ can be challenging.

Example in IDS: In the context of Intrusion Detection Systems (IDS), gradual Drift might be seen in the slow adaptation of attackers to new security measures, leading to a steady change in attack patterns as they transition from H to H' (Andresini, 2021).

Detection and Adaptation: Detecting gradual Drift may involve Sequential Analysis-Based Methods or Incremental Learning techniques. Adaptation might require continuous monitoring and incremental updates to the model to align with the evolving threat landscape and the transition from H to H' (Shyaa et al., 2023).

2. **Sudden Drift**: Sudden Drift encapsulates an abrupt and immediate alteration in the underlying concept, transitioning from H to H'. This type of concept drift is characterized by an instantaneous shift, catching models off guard and requiring rapid adaptation, as shown in Fig. 5(a), (Agrahari and Singh, 2022b).

Challenges.

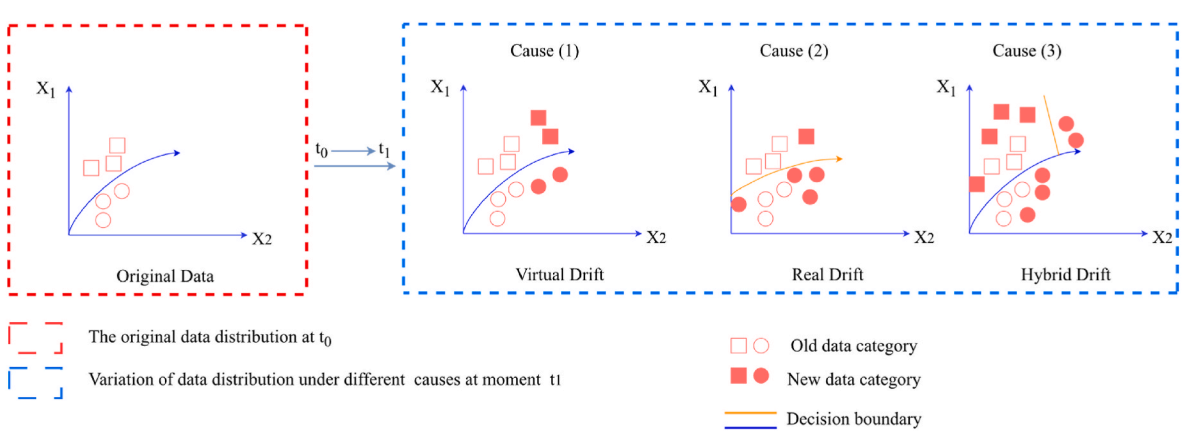

Fig. 4. The causes of concept drift (Xiang et al., 2023).

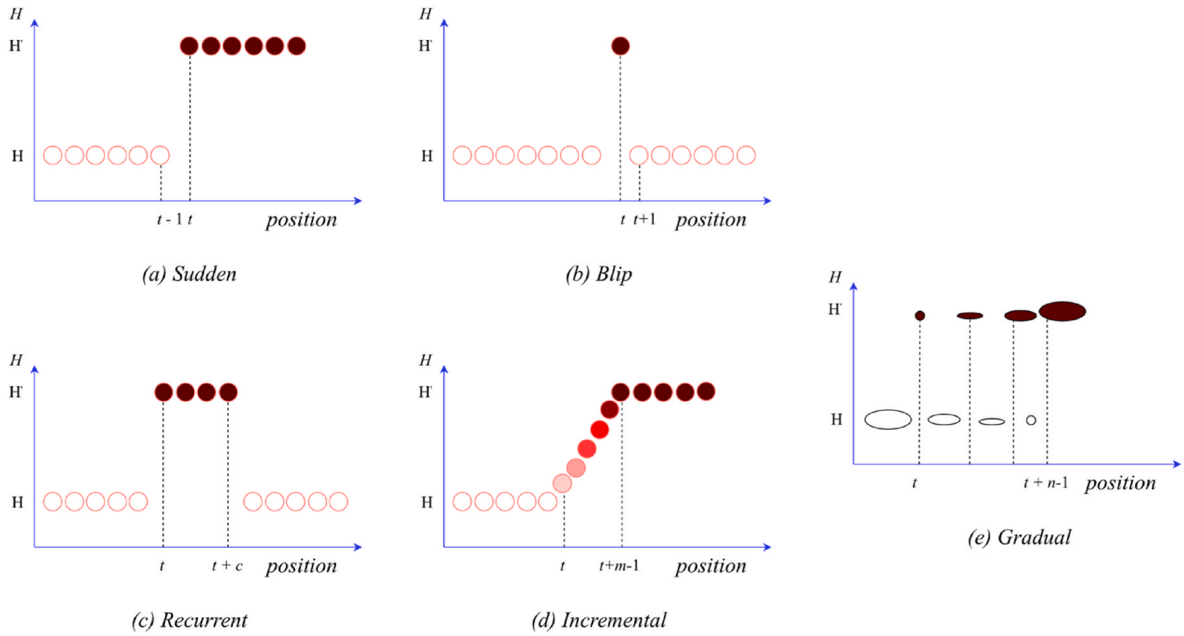

Fig: 5. The causes of Concept Drift (Xiang et al., 2023).

1. Abrupt Transitions: Sudden Drift poses the challenge of sudden and unforeseen changes from H to H′ that demand immediate model adjustments.
2. Unexpected Nature: The unexpected nature of sudden Drift makes it challenging to predict and prepare for, potentially causing models to make inaccurate predictions during the transition.
3. Real-time Adaptation: Rapid response and real-time adaptation are crucial for effectively handling the sudden transition from H to H′, requiring vigilant monitoring.

Example in IDS: sudden Drift could manifest as attackers swiftly changing their tactics to exploit new vulnerabilities, necessitating immediate updates to defence mechanisms as the concept shifts from H to H' (Kuppa and N. A. Le-Khac, 2022).

Detection and Adaptation: Detecting sudden Drift often involves anomaly detection techniques, where deviations from the expected pattern trigger alarms. Adaptation might necessitate ensemble methods or quick model retraining to respond promptly to the shifted concept from H to H' (Mahdi et al., 2021).

3. **Incremental Drift**: Incremental Drift involves a gradual but continuous evolution in the underlying concept, transitioning from H to H'. Concepts evolve and accumulate changes, making it imperative to adapt and update models progressively, as shown in Fig. 5(d) (Gama et al., 2014).

Challenges.

1. Continuous Evolution: Incremental Drift's ongoing and gradual nature requires continuous model adjustments to prevent performance degradation during the transition.
2. Accumulated Changes: The accumulation of minor shifts can lead to significant deviations if not addressed regularly as the transition progresses from H to H'.
3. Resource Efficiency: Ensuring efficient use of resources while updating models incrementally poses a challenge during the concept transition.

Example in IDS: Incremental Drift might manifest as attackers slowly refine their techniques to bypass existing defenses, necessitating gradual model updates as the concept transitions from H to H' (Kuppa and N. A. Le-Khac, 2022).

Detection and Adaptation: Detecting incremental Drift can involve Adaptive Window-Based Approaches or techniques that emphasize model stability. Adaptation might require online learning strategies, allowing the model to learn from incoming data while maintaining a balance with the past during the transition from H to H′ (Yang et al., 2021).

4. **Recurrent Drift**: Recurrent Drift signifies a cyclic or periodic change in the underlying concept, with concepts alternating between different states, such as between H and H′ in a recurring pattern, as shown in Fig. 5(c), (Ramírez-Gallego et al., 2017).

Challenges.

1. Cyclical Patterns: Recurrent Drift's cyclic behavior demands models capable of recognizing and adapting to repeating changes as the concept alternates between H and H'.
2. Pattern Duration: The varying duration of each cycle can complicate accurate prediction and timely adaptation during the recurrent transition.
3. Pattern Identification: Identifying the recurrence pattern and aligning models accordingly is complex during the concept transition from H to H'.

For example, in IDS, recurrent Drift might emerge as a seasonal pattern of attacks, requiring models to adjust to these periodic fluctuations as the concept alternates between H and H' (Soltani et al., 2024).

Detection and Adaptation: Detecting recurrent Drift may involve time-series analysis techniques or recurrent neural networks. Adaptation could entail capturing the cyclic behaviour and timing updates to coincide with anticipated shifts during the transition from H to H' (Suryawanshi et al., 2023).

5. **Blip Drift: Blip** Drift refers to a short-lived and sudden shift in the underlying concept, transitioning momentarily from H to H′ and quickly reverting to the original concept, as shown in Fig. 5 (b) (Kuncheva, 2008).

Challenges.

1. Transient Nature: Blip drift's brief and transient nature makes distinguishing from noise or outliers challenging during the short-lived transition from H to H'.
2. Impact Evaluation: Determining whether a blip requires adaptation or can be ignored involves careful impact assessment during the concept shift.
3. Model Overfitting: Overreacting to blip drifts might lead to unnecessary model changes that negatively affect performance during the transition from H to H'.

For example, in IDS, blip drift could occur as a momentary surge in malicious activities due to a specific event, demanding precise discrimination during the transition from H to H' (Soltani et al., 2024).

Detection and Adaptation: Detecting blip drift requires robust outlier detection methods or filters to identify short-lived disturbances. Adaptation might involve mechanisms that suppress quick model updates for minor blips while remaining vigilant for sustained shifts during the concept transition from H to H' (Guo et al., 2022).

2.3.1. Concept drift detectors

Conceptual conceptual handling of concept drift is presented in Fig. 6. Conceptual handling of concept drift is indeed presented in the Figure. In the initial phase, stream data is received, and a machine-learning model is constructed. This model serves the purpose of generating predictions. Concurrently, a subset of this stream data detects potential concept drift. Upon the identification of Drift, mechanisms are put in place: the model may invoke a forgetting process to shed outdated data while also potentially engaging in dynamic feature selection to recalibrate its predictive features in accordance with the latest data trends. The adaptation phase is critical, as it involves adjusting the model parameters or retraining the model with new data to cope with the observed changes. The culmination of this process is the interpretation of the concept drift, which is crucial for understanding the shifts in the data and the underlying factors that caused the change, thereby informing subsequent actions to refine the model's performance.

The modelled concept is not static; it changes over time, which can cause inconsistencies in the model. Therefore, it is essential to update the model regularly to ensure accuracy. As the distribution of data changes over time, errors may arise in the learning model. A detection mechanism tracks these errors in real-time to mitigate them. As provided in (Bayram et al., 2022) work, the concept drift detection algorithms are categorized into different types, as illustrated in Fig. 7.

2.3.1.1. Statistical process control (SPC). Statistical Process Control (SPC) is a criterion used to monitor the quality of the learning process. It does this by tracking the evolution of base learners' online error rates. Concept drift is assumed to have occurred when the model's performance degradation exceeds the level of significance test. Various performance-based methods, which can be found in the literature, use SPC to detect concept drift.

Various methods have been developed, in the scholarly landscape of Statistical Process Control (SPC) based concept drift detection, each with unique merits and limitations. The Drift Detection Method (DDM) described in the article (Gama et al., 2004) relies on monitoring error as a Bernoulli random variable. It is noted for its simplicity but faces limitations in handling slowly gradual changes. Early Drift Detection Method.

(EDDM), covered in the article (Baena-García et al., 2006) uses the distance between classification errors for detection and addresses slow drifts more efficiently. However, it struggles with the automatic determination of its parameters. The Reactive Drift Detection Method (RDDM) is outlined in the article (Barros et al., 2017) and is noted for its

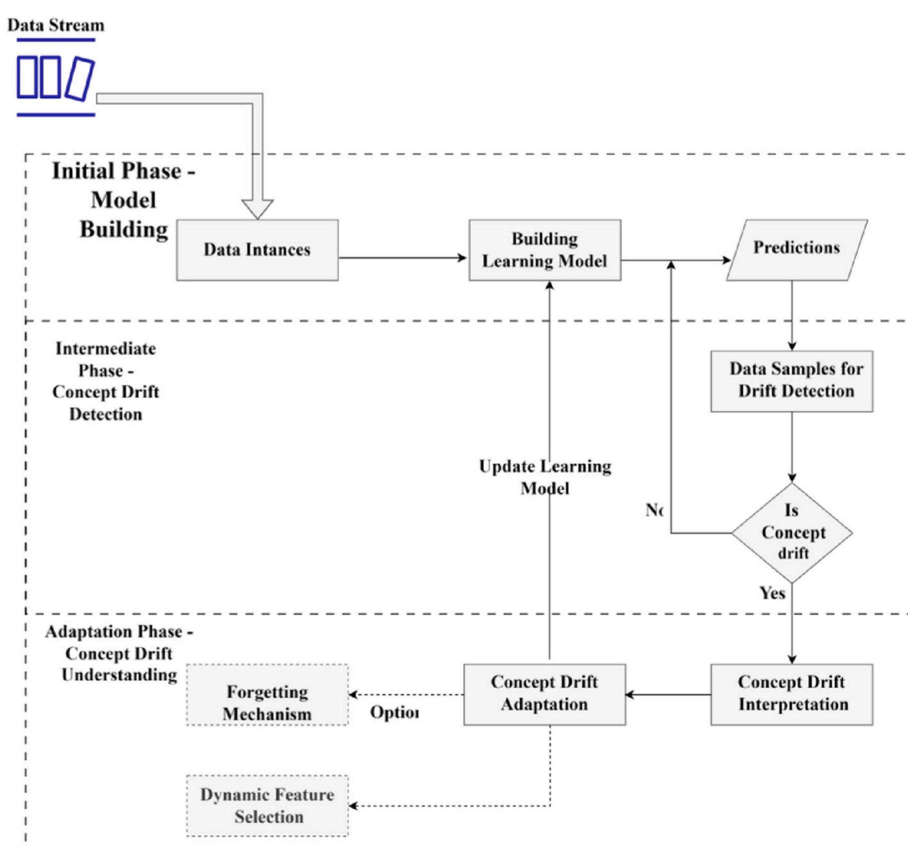

Fig. 6. General block diagram of concept drift detection (Agrahari and Singh, 2022b).

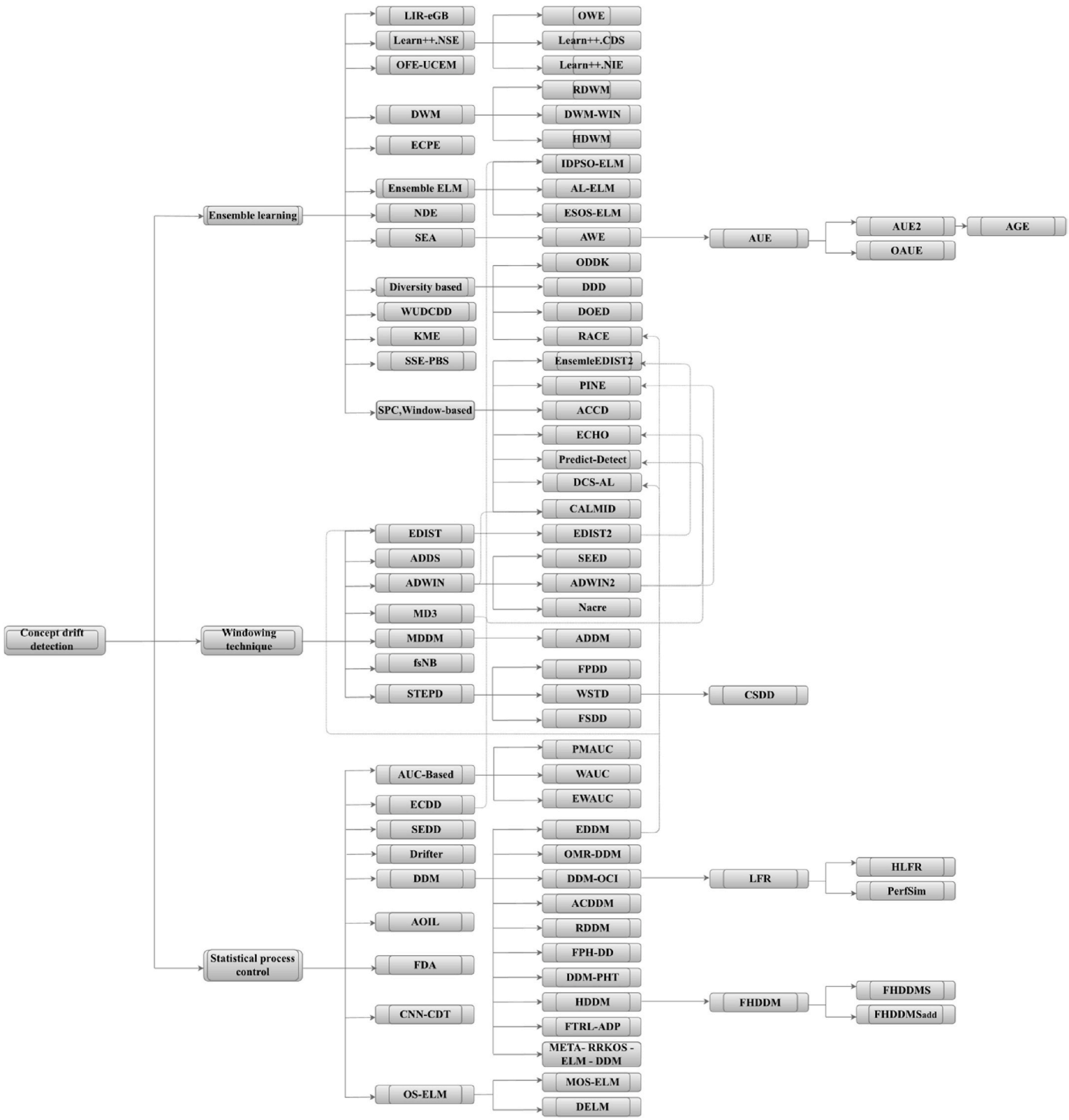

Fig. 7. Taxonomy of analysis performance-Oriented Drift detection methods (Bayram et al., 2022) .

higher global accuracy. However, its performance in imbalanced multi-classification problems is yet to be adequately investigated. HDDM-W, presented in the article (Frías-Blanco et al., 2015), generalizes the A-test and involves weighted averages, but its limitations are not explicitly mentioned. Fast Hoeffding Drift Detection Method (FHDDM) and its stacked extension (FHDDMS), featured in articles (Pesaranghader and Viktor, 2016; Pesaranghader et al., 2018a) offer timely and accurate concept drift detection but require parameter tuning. The article (Yan, 2020) introduces the Accurate Concept Drift Detection Method (ACDDM), which is effective at detecting true drifts but assumes consistent learning algorithms. Semi-supervised and unsupervised approaches, discussed in the article (Lughofer et al., 2016), offer competitive performance under different conditions but face certain limitations. Page-Hinckley Test (PHT) in the article (Mouss et al., 2004) utilizes statistical process control for real-time anomaly detection, albeit with the risk of false positives and negatives. A combined approach of DDM and PHT, covered in the article (Sakamoto et al., 2016), offers robust and versatile drift detection but demands parameter tuning.

DDM-OCI, described in the article (Wang et al., 2013), is designed explicitly for imbalanced data streams but can sometimes generate false alarms. The Linear Four Rates (LFR) Algorithm and its hierarchical variant (HLFR), featured in articles (Wang and Abraham, 2015; Yu et al., 2019), offer sensitivity to various types of Drift and adaptability but come with computational requirements and the need for parameter tuning. The PerfSim Algorithm, highlighted in the article (Antwi et al., 2012), employs a custom vector similarity approach and is sensitive to concept drift in imbalanced data but may require threshold tuning. The Fuzzy Drift Variance (FDA) method, introduced in the article (Song et al., 2020), can identify non-drift streams and simultaneous drift correlations but has limited abilities to identify delayed drift correlations. Adaptive Online Incremental Learning for evolving data streams (AOIL) from the article (Zhang et al., 2021) and Spectral Entropy Drift Detector (SEDD) from the article (Chikushi et al., 2021) introduce novel features like self-attention and spectral entropy-based detection but are limited in their focus on single-task streams and their handling of gradual datasets, respectively. As featured in the article (Oikarinen et al., 2021), Drifter is efficient in detecting virtual concept drift in regression tasks but makes certain distributional assumptions.

The Exponentially Weighted Moving Average for Concept Drift Detection (ECDD) methodology, referenced in (Ross et al., 2012), builds upon the traditional EWMA techniques discussed in (Yeh et al., 2008), tailoring them for monitoring fluctuations in classifier error rates. The ECDD computes the error rate and dynamic standard deviation at specific instances, signalling a concept drift when certain conditions are met (DIsabato and Roveri, 2019). have adapted Convolutional Neural Networks (CNNs) to incorporate Change Detection Tests (CDTs), focusing on classification error monitoring to detect concept drift using the CUSUM test (Page, 1954). A suite of AUC-based metrics, including Prequential Multi-Class AUC (PMAUC), Weighted AUC (WAUC), and Equal Weighted AUC (EWAUC), is proposed by (Wang and Minku, 2020). These metrics are crucial in multi-class imbalanced data scenarios, tracking their evolution over time. Furthermore, the application of Extreme Learning Machine (ELM) techniques for drift detection is explored by (Huang et al., 2006), particularly its online sequential variant (OS-ELM) (Liang et al., 2006). Yang et al. (2020) introduced a method identifying concept drift by analyzing variations in OS-ELM model output weights. The Dynamic Extreme Learning Machine (DELM) (Xu and Wang, 2017) augments ELM by adding drift detection that tracks learner performance degradation.

2.3.1.2. Windowing technique. In concept drift detection, multiple windowing techniques have been investigated and compared in the provided table. Adaptive Windowing (ADWIN) and its subsequent iteration, ADWIN2, as outlined in Article (Bifet and Gavaldà, 2007), are techniques that use sliding windows whose sizes are adjusted in real-time based on observed data changes; ADWIN2 improves explicitly computational efficiency. SEED (Stream Event Detection), presented in Article (Huang et al., 2014), contrasts two sub-windows within a single window for drift detection and is faster and more memory efficient than ADWIN2. STEPD (Statistical Test of Equal Proportions) in Article (Nishida and Yamauchi, 2007) employs a statistical test for drift detection, noted for its quick and accurate performance in sudden changes. The Wilcoxon Sum Rank Test Drift Detector (WSTD) in Article (Barros et al., 2018) uses the Wilcoxon rank sum test. It improves upon the detections of STEPD but is prone to false positives. Cosine Similarity Drift Detector (CSDD) in Article (Tetko et al., 2019) shows effectiveness in artificial datasets but performs less effectively with real-world data. McDiarmid Drift Detection Method (MDDM) in Article (Pesaranghader et al., 2018b) employs McDiarmid's inequality and is faster in detecting drifts. Adaptive sliding window-based Detection Method (ADDM) in Article (Du et al., 2014) uses information entropy and ranks high in accuracy but fourth in runtime. Margin Density Drift Detection (MD3) in Article (Sethi and Kantardzic, 2015) tracks density distribution changes in unlabeled samples and can degrade to a fully labeled approach if Drift is detected frequently. Fast Switch Naïve Bayes (fsNB) in Article (Liu et al., 2020) employs a Naïve Bayes classifier with a time window strategy and has performance affected by residual distribution and threshold choice. Finally, EDIST and EDIST2 in Articles (Khamassi and Sayed-Mouchaweh, 2014; Khamassi et al., 2015), respectively, are suited for detecting different types of drifts but are sensitive to threshold values and may require parameter tuning.

2.3.1.3. Ensemble learning. Ensemble-based concept drift detectors combine multiple diverse base learners to monitor overall performance accuracy. The article (Du et al., 2015) presents an ensemble of base detectors like DDM, EDDM, and EnDDM, enhancing robustness and accuracy while facing challenges in fusing diverse detector outputs. The Weighted Majority Algorithm in the article (Littlestone and Warmuth, 1994) relies on weighted voting and offers robustness in the presence of errors but struggles with slowly increasing statistical noise. Article (Street and Kim, 2001) Streaming Ensemble Algorithm (SEA) achieves efficiency and quick adaptability but re-examines individual points multiple times when using the C4.5 algorithm. The Accuracy Weighted Ensemble (AWE) in the article (Wang et al., 2003) adapts well to underlying concept drifts but struggles to determine outdated examples. Article (Brzeziński and Stefanowski, 2011) describes the Accuracy Updated Ensemble (AUE), which enhances efficiency and handles concept drifts effectively, albeit with issues concerning data block size. The article (Brzezinski and Stefanowski, 2014) highlights the Online Accuracy Updated Ensemble (OAUE) as fast and cost-effective but falls short in handling rare abrupt changes. The article's Accuracy and Growth Rate Updated Ensemble (AGE) (Liao and Dai, 2014) adapts quickly to current concepts but may reduce accuracy during stable periods. Article (Mejri et al., 2013) DWM-WIN improves predictions over single classifiers but does not account for expert contributions during training. The HDWM method in the article (Idrees et al., 2020) adjusts to concept drifts by dynamically changing base learner weights but faces potential accuracy reduction due to the new model's unreliability. Article (Sidhu and Bhatia, 2019) RDWM efficiently handles recurring drifts but relies solely on the Ensemble with better generalization accuracy for its predictions.

For example, the Learn++ algorithm (Polikar et al., 2001) employs an ensemble of weak classifiers and dynamically adjusts the weights of training examples to focus on instances that the current Ensemble misclassifies. This approach is particularly suited for incremental learning and accommodates new classes, although selecting its hyperparameters remains an ad-hoc process. Extensions of Learn++ (Ditzler and Polikar, 2013) such as Learn++.CDS and Learn++.NIE further specialize in addressing imbalanced and drifting data, incorporating time-adjusted accuracy metrics and strategic data sampling methods.

The Online Weighted Ensemble (OWE) (Gomes Soares and Araújo, 2015) focuses on handling multiple types of drifts, from abrupt to gradual and from global to local, while the Diversity for Dealing with Drifts (DDD) (Minku and Yao, 2012) method maintains ensembles at varying levels of diversity to better cope with concept drifts in online learning scenarios. Other algorithms like DOED (Sidhu and Bhatia, 2015) and RACE (Museba et al., 2021) are optimized for high-speed drifts and recurring concepts, respectively, although they come with computational or memory overheads.

On the other hand, PINE (Ang et al., 2013) is designed to effectively handle asynchronous concept drifts but struggles with scalability issues. The CALMID (W. Liu, 2021b) framework integrates multiple components, from an ensemble classifier to a drift detector and sliding windows, for handling multi-class imbalanced streaming data. However, due to its high computational demands, it is unsuitable for real-time applications.

Associative Classification over Concept Drifting Data Streams (ACCD) (Waiyamai et al., 2014) evaluates ensemble classifier accuracy against a statistically lower accuracy bound for drift identification. Ensemble EDIST2 (Khamassi et al., 2019) integrates EDIST2 within its ensemble framework for ongoing learner performance assessment. The predict-detect streaming framework (Sethi and Kantardzic, 2018) inspired by MD3 detects adversarial drifts in unlabeled streams by benchmarking expected disagreement and deviation within an ensemble. A marked increase in disagreement signals drift.

Efficient Concept Drift and Concept Evolution Handling over Stream Data (ECHO) (Haque et al., 2016) adopts a semi-supervised approach using a sliding window and CUSUM test to monitor significant classifier confidence changes, indicating concept drift. Khezri et al. (2021) propose a Performance-Based Selection (PBS) metric in a semi-supervised setting, evaluating model performance through pseudo-accuracy and energy regularization, enhancing assessment accuracy in dynamic data environments.

The ESOS-ELM (Mirza et al., 2015) framework addresses class imbalance in concept-drifting data streams. However, it requires tuning and adjustments, similar to the IDPSO-ELM-B & IDPSO-ELM-S (Oliveira et al., 2017) methods that use Extreme Learning Machines and are robust to false positives. Advancements include the Number and Distance of Errors (NDE) (Dehghan et al., 2016), which detects concept drift through error pattern analysis. The Knowledge-maximized Ensemble (KME) (Ren et al., 2018) incorporates a TEST drift detector within a sliding window to assess classification accuracy. The Enhanced Concept Profiling Framework (ECPF) (Anderson et al., 2019) adopts a meta-learning approach, focusing on behavioral changes in learners for drift detection. A novel pruning criterion, the Loss Improvement Ratio (LIR), introduced by Wang et al., evaluates and strategically prunes outdated learners.

The Weighted classification and Update algorithm of the Data stream based on Concept Drift Detection (WUDCDD) (Zhang and Chen, 2019) integrates classification with dual detection mechanisms, showcasing superior accuracy but necessitating further research in update strategies.

Each method offers unique advantages and drawbacks, often tailored for specific types of data drift, imbalances, or computational constraints. Thus, choosing an appropriate algorithm depends on the particular application scenario.

2.3.2. High dimensionality

In the context of Intrusion Detection Systems (IDS), high dimensionality presents a substantial challenge, particularly when interlaced with concept drift, a phenomenon where the statistical properties of data change over time. This section investigates the complex interplay between the abundance of features in a dataset and the agility IDS requires to adapt to evolving data trends. We will examine strategies to manage high dimensionality, including dimensionality reduction techniques like Principal Component Analysis (PCA) and Fisher Discriminant Analysis, and their effectiveness in a landscape where feature relevance is in constant flux. The focus is to assess these methodologies critically, identify existing shortcomings, and suggest directions for future research that could enhance the adaptability of IDS in high-dimensional, dynamic settings.

2.3.2.1. Dimension reduction through PCA. Principal Component Analysis (PCA) is a fundamental statistical approach in data science, precious for managing the complexities associated with multi-dimensional data sets. At its core, PCA aims to distill data that originally had many dimensions into a more manageable number of dimensions, thereby simplifying analysis and enhancing interpretability (Moore, 1981; Abdi and Williams, 2010); this method is particularly adept at uncovering patterns within the data and restructuring it to emphasize critical similarities and distinctions. PCA reduces data dimensions while preserving essential relationships and patterns, making it an indispensable tool in handling large and complex datasets (Ayesha et al., 2020).

PCA's utility is most evident when working with extensive datasets where analyzing all variables simultaneously can be challenging. It effectively condenses the data by transforming the original variables into a new set of uncorrelated variables, known as principal components. These components are arranged in a manner where the initial few retain a significant portion of the variability found in the full set of original variables (Kasun et al., 2016; Keerthi Vasan and Surendiran, 2016; Nasution et al., 2018). The primary objective of PCA is to identify a k-dimensional subspace where these principal components are situated. The first principal component is particularly vital as it encompasses the most significant variance, thereby playing a pivotal role in elucidating the data structure. PCA is most beneficial when there is a collinear relationship among the variables in the dataset. A critical step in the PCA process is choosing the optimal number of principal components, denoted by k, ensuring an effective balance between dimensionality reduction and preserving the original data's integrity.

In this Table 5, (Oo and Thein, 2022), the input encompasses the training dataset labeled D, the desired number of principal components k, and the specified number of feature variables m. The anticipated output is a set A comprising k components with m feature variables. The process initiates with standardizing dataset D. This is followed by determining the mean-centred matrix, M_c, and its subsequent covariance matrix. The Eigenvalue decomposition of this covariance matrix yields X_c. After extracting the associated Eigenvectors and Eigenvalues and ordering the latter in descending fashion, the Eigenvector tied to the highest Eigenvalue is selected. A matrix is then constructed using these chosen Eigenvectors. The algorithm culminates in obtaining the principal components of X and returning F, although it should be highlighted that F is not explicitly defined in the given pseudocode.

2.3.2.2. PCA in an offline or batch setting. The core concept of Principal Component Analysis (PCA) is centred on capturing the maximum variance in a dataset using a constrained number of linear descriptors. This methodology is particularly effective for datasets characterized by high dimensionality (Greenacre et al., 2022). PCA operates by projecting these datasets onto a smaller number of dimensions. In this process, ensuring that the variance along these newly formed axes is maximized is crucial, thereby encapsulating the most significant aspects of the data. A key characteristic of these new dimensions, or principal components, is their mutual orthogonality, which ensures that each component represents a distinct aspect of the data's variance without redundancy (Migenda et al., 2021).

Fig. 8(a) illustrates a three-dimensional data distribution. The distribution's three axes through PCA are ascertained based on the descending variances of projected data. The initial principal component serves as a linear descriptor, capturing the majority of the data's variance. Once this primary component's value is established, subsequent principal components are determined to account for the remaining significant variance. Defining a data point using only the first two principal components might be adequate for the scenario depicted, as shown in Fig. 8(b). This underscores PCA's utility in reducing dimensionality.

Traditional batch-PCA or offline PCA (Jollife and Cadima, 2016; Tharwat, 2016) uses an orthonormal transformation to convert a potentially interrelated dataset into a collection of linearly independent variables. Patterns with high dimensionality of n dimensions can be represented in a subspace with reduced m dimensions, transitioning from IR^n to IR^m. A PCA model depicts this subspace using m principal components, where m is always less than or equal to n. Moving forward, the $n \times m$ matrix W represents the normalized eigenvectors w_i (for $= 1, \ldots, m$) of the covariance matrix of the data. Each column corresponds to a vector. These eigenvectors align with the principal components. The data distribution's projection variance onto the i^{th} principal component w_i is equivalent to the eigenvalue λ_i. All eigenvalues λ_i are arranged in descending order within a diagonal matrix Λ that measures $m \times m$. Estimates for the remaining eigenvalues in the $n - m$ minor eigendirections can be sourced from the residual variance σ^2. In addition, a centre vector c belonging to IR^n is essential for centring the PCA. As a result, multivariate data can be characterized solely by the matrices W and Λ, the vector c, and σ^2.

Table 5
DR_PCA (Dimension Reduction using PCA).

Input:
- D: Training dataset
- k: Number of principal components to retain
- m: Chosen number of feature variables

Output:
- A: Set of k components, each having m feature variables

Procedure:
1 Standardize the training subset D.
2 Derive the mean-centred matrix, denoted as M_c.
3 Compute the covariance matrix, represented as Cov $(M_c) = M_c' \times M_c$.
4 Perform Eigenvalue decomposition on the covariance matrix to obtain X_c.
5 Extract the Eigenvectors and Eigenvalues from the decomposition of X_c.
6 Organize the Eigenvalues in a descending sequence.
7 Opt for the Eigenvector corresponding to the highest Eigenvalue.
8 Form a matrix by projecting the selected Eigenvectors.
9 Acquire the k principal components of $X : A = (a_1, a_2, \ldots, a_k)$.
10 Return F.

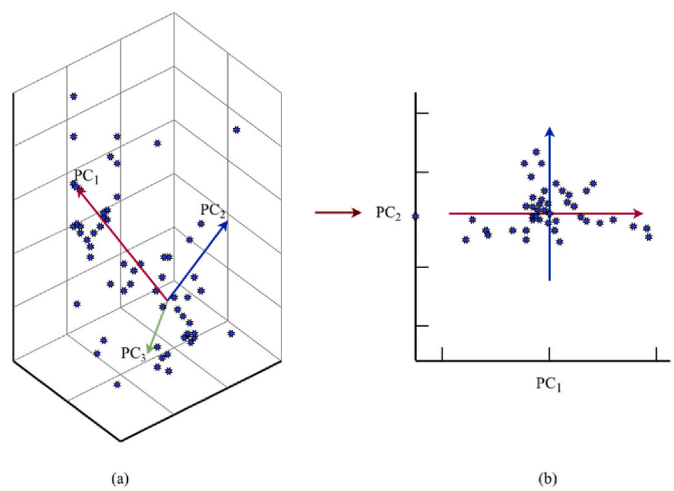

Fig. 8. Illustrates the PCA-based dimensionality reduction process: Part (a) shows data in a three-dimensional space with principal components, while part (b) depicts the data's projection onto a two-dimensional plane using PC1 and PC2.

2.3.2.3. PCA in a streaming setting. In PCA applications within streaming contexts, the environment is characterized by continuous data points arriving sequentially over time. In such settings, the parameters that define the PCA subspace are constantly in flux, necessitating regular updates (Balzano et al., 2018). As the data evolves, significant changes can be observed either in the covariance matrix or the subspace itself, necessitating ongoing adjustments to the PCA model. This scenario underscores the importance of continuously monitoring and adapting the PCA model to accurately reflect the latest data characteristics (Migenda et al., 2021). In response to these challenges, online PCA techniques have been developed. These methods are uniquely designed to update their parameters regularly and autonomously without the need to access the entire backlog of historical data. This approach is beneficial in environments where data is continually changing. Key algorithms within the online PCA framework include incremental PCA (Hall et al., 2013), which integrates new data points while preserving the underlying structure identified in previous data; incremental SVD (Brand, 2006), an efficient method for updating decompositions with new incoming data; and PCA models based on neural networks (Oja, 1982), which use adaptive learning algorithms to refine the PCA analysis in light of new data constantly. These online PCA methods are essential in dynamic data environments, offering the flexibility and adaptability needed for real-time data analysis and interpretation.

2.3.2.4. Fisher discriminant analysis. Fisher's Discriminant Analysis (FDA) is a traditional technique for projecting a data matrix to achieve maximal discrimination, and it has proven effective in classification performance and dimension reduction (Fukui et al., 2023). This method is employed to create a discriminant space, referred to as H, which efficiently differentiates between various classes (FISHER, 1936; Nasution et al., 2018). Maximizing the Fisher criterion based on the data projected onto H determines the optimal discriminant space H. This criterion is formulated using two key components: the within-class covariance matrix and the between-class covariance matrix. Considering that there are C distinct classes, each with its dataset, the FDA optimally separates them in the identified discriminant space H.

$\{\mathbf{x}_i^c\}_{i=1}^{n_e}(c = 1, \ldots, C)$, the within-class covariance matrix $\Sigma_W \in \mathbb{R}^{L\times L}$ is defined as in Equation (3)

$$\Sigma_W = \sum_{c=1}^{C} p(c)\left(\frac{1}{n_c}\sum_{i=1}^{n_c} \left(\mathrm{x}_i^c - \mathrm{m}\right)\left(\,^{c}_{usi} - \mathrm{m}_c\right)^T\right) \tag{3}$$

Where $p(c)$ represents the prior probability of class c, with n_c and m_c denoting the sample count and mean vector of class c, respectively. The matrix that defines the between-class covariance, Σ_B, is in $R^{(L\times L)}$ and is described in Equation (4)

$$\begin{aligned}\Sigma_B \quad &= \sum_{c=1}^{C} p(c)(\mathrm{m}_c - \mathrm{m})(\mathrm{m}_c - \mathrm{m})^T \\ &= \sum_{i=1}^{C-1}\sum_{j=i+1}^{C} p(i)p(j)\left(\mathrm{m}_i - \mathrm{m}_j\right)\left(\mathrm{m}_i - \mathrm{m}_j\right)^T\end{aligned} \tag{4}$$

In this context, $p(i)$ and $p(j)$ represent the prior probabilities for the i-th and j-th classes, respectively, while m.

Signifies the mean vector encompassing all the classes. The Fisher criterion, denoted as $f(d)$, for data projected onto a one-dimensional subspace governed by the vector d is described in Equation (5)

$$f(\mathrm{d}) = \frac{\mathrm{d}^T\Sigma_B\mathrm{d}}{\mathrm{d}^T\Sigma_W\mathrm{d}}, \tag{5}$$

Wherein the vector d, which optimizes the function f, can be deduced by addressing the generalized eigenvalue issue.

$$\Sigma_B\mathrm{d} = \lambda\Sigma_W\mathrm{d}. \tag{6}$$

The discriminant space H is defined by $C-1$ eigenvectors, $d_{ii=1}^{C-1}$, and is associated with the top $C-1$ eigenvalues stemming from the aforementioned eigenvalue equation.

2.3.2.5. Dimension reduction with IG. Information Gain (IG) stands out as an advanced method for feature selection, particularly valuable in simplifying datasets characterized by high dimensionality. Central to refining experimental datasets, IG focuses on evaluating the significance of each feature in the dataset. The primary objective is to analyze and prioritize features based on their informational value, facilitating a more efficient analysis process (M. Li et al., 2020a). The core principle of this method involves ranking attributes by computing their Gain Ratio, a metric that quantifies the information provided by each feature. The feature with the highest Gain Ratio is considered the most significant. This prioritization helps select the most informative attributes, potentially reducing the risk of overfitting. IG effectively identifies the attribute that contributes most significantly to the prediction or classification task. The procedure for dimensionality reduction using IG is delineated in Table 6 (Oo and Thein, 2022).

The entropy for a specific subset of the dataset is defined in Equation (7).

$$\text{Entropy}\ (S_i) = -\sum_{c=1}^{d} q_c \times \log q_c \tag{7}$$

Here, S_i represents a subset of the dataset S, d indicates the number of distinct classes within S_i, and q_c signifies the probability of occurrences in S_i that pertain to class c.

The entropy for each variable y_{ij} of the subset S_i is expressed in Equation (8).

$$\text{Entropy}\ \left(y_{ij}\right) = \sum_{v \in V\left(y_{ij}\right)}^{d} \frac{\left|S_{(v,i)}\right|}{|S_i|} \times \text{Entropy}\ \left(v\left(y_{ij}\right)\right) \tag{8}$$

In this Equation, y_{ij} is the j th input feature variable of the subset S_i, $v\left(y_{ij}\right)$ is the Ensemble of potential values for each input variable, and $S_{(v,i)}$ stands for the sample subset of dataset S_i.

The information split of each input variable is elaborated in Equation (9).

$$I\left(y_{ij}\right) = \sum_{c=1}^{m} -q_{(c,j)} \times \log\ \left(q_{(c,j)}\right) \tag{9}$$

Here, m is the count of distinct classes of y_{ij}, and $q_{(c,j)}$ signifies the probability of occurrences of y_{ij} that fall under class c.

The information gain value for each feature variable is defined in Equation (10).

$$G\left(y_{ij}\right) = \text{Entropy}\ (S_i) - \text{Entropy}\ \left(y_{ij}\right) = \text{Entropy}(S_i) - \sum_{v \in V\left(y_{ij}\right)}^{d} \frac{\left|S_{(v,i)}\right|}{|S_i|} \times \text{Entropy}\ \left(v\left(y_{ij}\right)\right) \tag{10}$$

Table 6
Dimensionality reduction via IG (DR_IG).

Input:
- S: the training dataset
- k: desired number of features to select
- m: total number of feature variables in S.

Output:
- F: a collection of m significant features from S.

Procedure:
1 Normalize the training subset S_i.
2 Compute the entropy of target features.
3 For each variable y_{ij} in training dataset S:
- Compute the entropy for y_{ij}.
- Determine the splitting node information for y ou.
- Evaluate the information gained for y_{ij}.
- Determine the gain ratio for y ou.

4 End loop.
5 Assess the feature significance for each variable y_{ij}.
6 Rank the values of Fe-imp $\left(y_{ij}\right)$ in descending order.
7 Select the top k features and add them to F.
8 Output F.

The execution time for SRF is diminished by excluding features that do not influence the prediction strength. DR_IG functions to minimize the training samples, which could undermine the predictor's efficacy. More precisely, an attribute's entropy determines the variable's gain ratio and gauges the feature's significance. Hence, accounting for the feature's significance in SRF is highly beneficial for processing large, high-dimensional datasets.

3. Literature review

In our comprehensive exploration of academic literature on concept and feature drift within intrusion detection systems, we employed a strategic and methodical approach to gather relevant research papers. As depicted in Fig. 9, this rigorous process involved an extensive search across eminent digital libraries such as Web of Science, Science Direct, Springer, IEEE Xplore, and Google Scholar, covering literature from 2019 to 2024. A meticulously crafted query guided our search:

(("Concept Drift" OR "Real Drift") OR ("Feature Drift" OR "Virtual Drift" OR "Covariate Shift") AND ("Intrusion Detection" OR "Intrusion Detection System" OR "IDS" OR "Network Security")) AND ("Data Stream" OR "Stream Data" OR "Online Learning") AND ("Machine Learning" OR "ML" OR "Deep Learning" OR "DL")).

The initial retrieval of 1649 articles underwent a two-stage screening process, detailed in Tables 7 and 8, resulting in a focused set of 71 highly relevant studies. This selective approach was underpinned by a robust quality assessment, outlined in Table 9, ensuring the inclusion of studies that were not only pertinent but also methodologically sound and of high academic integrity.

Transitioning from this carefully structured methodology to the content of our review, we delve into the ever-evolving domain of artificial intelligence, mainly focusing on the adaptability of systems in the dynamic field of intrusion detection. Our survey systematically dissects the current state-of-the-art in Concept Drift-Aware IDS, identifies key challenges, and elucidates the frontiers of research in managing dynamic data streams. Section 1, "Concept Drift Aware IDS Methods," begins the exploration by scrutinizing methods through which IDS adapt to concept drift, reflecting the intrinsic volatility of the cybersecurity

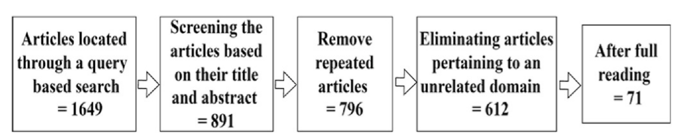

Fig: 9. Process of selecting studies.

Table 7
Inclusion criteria.

- Studies primarily address concept or feature drift within intrusion detection systems or network security.
- Research utilizing machine learning (ML) or deep learning (DL) methodologies to tackle concept and feature drift issues in intrusion detection or network security contexts.
- Investigations focus on concept and feature drift in data stream management or online learning environments.
- Publications limited to English-language journal articles known for their credibility and academic rigour.
- Research articles published in the timeframe from 2019 to 2024, ensuring contemporary relevance.

Table 8
Exclusion criteria.

Exclusion Criteria
Research papers that do not focus on concept or feature drift within The domains of intrusion detection systems, network security, data streams, and online learning are excluded.

Table 9
Checklist for assessing quality.

1. Have the studies focused on dealing with the problem of concept and feature drift in intrusion detection systems, network security, data streams, or online learning?
2. Does the study use ML/DL methods?
3. Does the research use a clear solution or adequate methodology?

landscape. Subsequent sections broaden the discussion to encompass various domains challenged by concept drift, assess dynamic feature selection methods, explore the integration of reinforcement learning, and finally, in Section 5, address the conjunction of concept drift with issues like class imbalance and high dimensionality. Through this structured examination, our survey not only chronicles advancements in system adaptability but also fortifies the foundation for future research endeavors in this critical and evolving Area of cybersecurity.

3.1. Concept drift aware IDS methods

The literature has showcased a plethora of methodologies and models aimed at identifying attacks in streamed data. (Seth et al., 2024) introduced an Adaptive Random Forest (ARF) model for intrusion detection within data streams. Utilizing Hoeffding's bounds combined with a moving average-test, the model effectively identifies sudden and gradual concept drifts. Comprehensive evaluations on the CIC-IDS 2018 dataset demonstrate the model's robustness and adaptability in dynamic environments. Nonetheless, it is important to note that this approach does not address incremental, recurring, or blip drifts. A unique model named INSOMNIA was introduced by (Andresini et al., 2021). This intrusion mitigation strategy persistently refines its ML model in response to CD. It employs active learning and label estimation to minimize update latency and labeling overhead. Moreover, it employs explainable AI for a more straightforward interpretation of model responses. Nonetheless, it does not cater to specific CD variants like recurrent, blip, and combined drifts (Uccello et al., 2024). developed a real-time concept drift detection method within a SIEM framework, using Page-Hinkley Test and ADWIN detectors to adapt to evolving attack patterns. This approach effectively detects sudden and gradual drifts, enabling timely updates to security rules. Evaluations on CIC-IDS2017 and CSE-CIC-IDS2018 datasets confirm its adaptability to emerging threats. However, it does not address incremental and recurring drifts.

The method presented by (Folino et al., 2020) () highlights an ensemble strategy for online IDS. This model is periodically refined using an incremental stream-based learning technique, leveraging genetic programming (GP) for combining functions. However, it only partially supports incremental learning, potentially hindering its adaptability to certain drift types like gradual drift.

In the study (Shyaa et al., 2023), the authors introduce a modified GPC framework with three enhancements: replacing traditional classifiers with OSELM variants, including FA-OSELM and KP-OSELM, integrating data balancing and classifier updating modules, and developing three GPC variations (GPC-KOS, GPC-FOS, and GPC-OS). This novel data stream classification system innovatively addresses concept drift (CD) types. Experimental results show GPC-KOS and GPC-FOS significantly outperforming the standard GPC and other advanced methods.

The article (Soltani et al., 2024) presents a framework for adapting deep learning models to dynamic attack and benign traffic patterns in online, adaptable IDSes. It combines continuous deep anomaly detection with federated learning to tackle traffic concept drift, including new attacks and changing benign behaviours. The framework also features sequential packet labeling for network flows, offering updated attack probability scores through ongoing packet analysis.

(Rajeswari et al., 2022) developed an IDS utilizing a novel SVM approach tailored for complex data environments. This system, equipped with an adaptive mechanism for windowing, capitalizes on enhanced data preprocessing and real-time analysis to pinpoint and adapt to evolving security threats. When tested against prominent datasets like KDD Cup 1999; CICIDS 2017, this model demonstrated a notable improvement in detection metrics, surpassing traditional IDS methods (Alqahtani, 2024). introduced an Incremental Hybrid Adaptive IDS for Software Defined Networks (SDNs) to detect stealth attacks. The system combines Adaptive Random Forest (ARF) for known attacks and Adaptive One-Class SVM (OC-SVM) for unknown attacks, adapting to sudden and gradual concept drifts. This dual-layer approach ensures timely and accurate detection.

The paper (Alsuwat et al., 2023) introduces a new method for detecting concept drift and DDoS attacks in networks to improve data representation and predict cybersecurity threats more effectively. It utilizes secure adaptive windowing and website data authentication protocol (SAW_WDA) to analyze the network's intrusion detection capabilities.

Another noteworthy study (Abdel Wahab, 2022) discussed a drift detection mechanism using PCA to track feature variance changes in intrusion data streams combined with an evolving deep neural network (Adnan et al., 2020). unveiled a hyper-heuristic strategy for CD management in IDS, intertwining genetic optimization, neural structures, and dual-mode stream clustering. However, its efficacy against various CD types was not assessed.

The study (Yang et al., 2021) introduces the Performance Weighted Probability Averaging Ensemble (PWPAE), a novel method for evaluating IoT data streams. This technique is notable for its adaptive weighting system, which adjusts the influence of individual base learners in real-time according to their ongoing performance, as opposed to relying on static weight assignments (Seth et al., 2021) proposed a dynamic online IDS model that harnesses stream-based learning for adapting to real-world CD. This model combines an Adaptive Random Forest classifier with an ADWIN detector for drift detection. However, its performance against diverse CD types remains unevaluated. (Martindale et al., 2020), conducted an evaluative study on three online heterogeneous ensemble models for IDS, assessing their response to CD, accuracy, and computational complexity. Their findings revealed the superior performance of the ensemble merging an Adaptive Random Forest of Hoeffding Trees with a Hoeffding Adaptive Tree in CD management. However, its performance against different CD types was not examined.

A distinct methodology for predicting CD in real-time for social network attack data streams was proposed by (mahmodi et al., 2020) . This technique amalgamates online learning mechanisms and expert consensus to pinpoint anomalies. In (Qiao et al., 2022), Qiao and colleagues investigate the impact of concept drift on identifying botnet intrusions within IoT frameworks. The study employs concept drift examination on the Bot-IoT dataset, which encompasses authentic and emulated IoT traffic, to discern various cyber-attack patterns.

(Yang and Shami, 2023) described a comprehensive network analytics model for industrial IoT systems' CD adaptation. This model encompasses dynamic feature selection, dynamic model training, and a probability-based ensemble model (dos Santos et al., 2022). suggested an innovative IDS model anchored in reinforcement learning. It consists of two main facets, designed for prolonged operation with infrequent updates. One focuses on long-term ML learning, and the other on model refinement using Transfer Learning and a sliding window framework (Jain and Kaur, 2021). put forth a distributed ML-ensemble method to identify CD in network traffic and network attacks. Their model comprises three components: employing Random Forest and Logistic Regression as foundational learners, using Support Vector Machine (SVM) as an advanced learner, and implementing a sliding window K-means clustering. However, the efficacy of this model against different CD types remains unreported.

(Kuppa and N.-A. Le-Khac, 2022) put forward a CD detection method tailored for real-world security scenarios. This technique relies on an embedding network with a contrastive loss function and a unique distance metric for NCM classifier training. However, it does not prioritize adaptation and struggles with incremental drift detection. In their study (Xu et al., 2023), L. Xu and colleagues developed ADTCD, an adaptive anomaly detection model countering concept drift via knowledge distillation. This approach, focusing on updating only the student model, minimizes delay and incorporates an algorithm that adjusts parameters based on new data, modifying weights after analyzing incoming samples. The study (Werner et al., 2021) introduces CLEAR, a real-time intrusion event aggregation system through concept learning. Without requiring pre-existing data, CLEAR dynamically generates and refines 'concepts' or clusters of statistically similar alerts as they arise. These evolving concepts then help quickly and accurately cluster new alerts. Finally, the article (Shao et al., 2021) presents an adaptive online learning strategy for real-time IoT botnet detection, highlighting its effectiveness in adapting to traffic pattern changes and concept drift. Incorporating online ensemble learning, the strategy shows significant performance improvements.

Table 10 presents a synthesized view of the models documented in scholarly works that pertain to predicting data streams.

3.2. Concept drift aware methods

The literature proposes various methodologies to address the challenges of detecting and managing concept drift (CD) in data streams.

The article (Chen and Dai, 2024) introduced Continuous Kernel Learning with Ensemble Methods and FHDDM for adapting to concept drift in data streams. This method targets sudden, gradual, and incremental drifts by dynamically adjusting to data changes, maintaining classification performance through ensemble learning. However, it does not address recurring or blip drifts. The article (Karimian and Beigy, 2023) introduces CDDA, a domain adaptation method for handling concept drift in data stream prediction, optimizing processing time and memory efficiency. It proposes two CDDA variants for effective data transfer and demonstrates their efficacy through extensive tests on various data streams (Chen et al., 2024). proposed the DLVSW-CDTD algorithm, which employs a dual-layer variable sliding window mechanism to dynamically adjust to different types of concept drift in data streams. This approach enhances the timely detection and adaptation to sudden and gradual drifts by capturing both local and global data patterns. While the algorithm demonstrates robust performance and efficiency, it may require further development to manage high-dimensional data and integrate advanced machine-learning techniques.

A notable hybrid block-based ensemble was detailed in (Mahdi et al., 2021) , offering a multi-class classification framework for dynamic data streams. This framework aims to amalgamate the benefits of an online drift detector for a K-class challenge with block-based weighting principles. However, its versatility against all drift types remains unverified, and comparative evaluations with other adaptation methods are lacking.

The study (Suryawanshi et al., 2023) introduces a novel recurrent neural network framework for effectively handling concept drift in data stream classification. This approach incorporates a gated recurrent unit, enhancing adaptability through elastic weight consolidation. It stands out by its dynamic window size adjustment, which responds to prediction accuracy, showing significant performance improvements on real-world and synthetic datasets compared to conventional models. In (Gâlmeanu and Andonie, 2021), an Incremental-Decremental SVM CD model, termed AIDSVM, was introduced. This model offers a significant advantage in speed over traditional SVMs by eliminating the need for continuous retraining and enhancing accuracy by timely removal of irrelevant samples based on the Hoeffding test. However, incremental construction often exhibits a slower build rate. The Recurrent Adaptive Classifier Ensemble (RACE) was unveiled by (Museba et al., 2021) and devised to tackle recurring concepts. The algorithm integrates a drift detector to ascertain the occurrence of a drift. Despite its promising prediction accuracy for non-stationary data, the RACE approach can be resource-intensive, demanding significant memory allocation, thus potentially delaying the convergence to recurring concepts.

(Wu et al., 2022), introduced Probabilistic Exact Adaptive Random Forest with Lossy Counting (PEARL). It focuses on substituting drifting trees with historically relevant trees. While it showcases precision in tree replacement, the method may exhibit biases due to the lack of ensemble learning and could overlook specific drift categories. NACRE, detailed in Article (Wu et al., 2021), offers proactive drift detection but necessitates a balance between drift detector sensitivity and sequence prediction signals.

In (Chiu and Minku, 2022), a Diversity framework was articulated, capitalizing on model space clustering to craft a diverse ensemble and recognize recurring concepts. Despite its innovative approach, the model lacks provisions for balanced learning. A heterogeneous adaptive ensemble model for stream data classification was discussed in (Sarnovsky and Kolarik, 2021). Comprising diverse essential learners, the model emphasizes ensemble diversity and individual performance. However, it sometimes overlooks specific drift nuances.

Kappa Updated Ensemble (KUE), mentioned in (Cano and Krawczyk, 2020), synergizes online and block-based strategies for CD management. KUE employs the Kappa statistic and focuses on dynamic base classifier weighting and selection. However, its application predominantly remains restricted to frequent drift scenarios. The Broad Ensemble Learning System (BELS), derived from the original BELS and elaborated in (Bakhshi et al., 2023), incorporates a dynamic output ensemble layer. However, it struggles with semi-supervised learning scenarios.

In the stud y (Wang et al., 2022b), the researchers introduced a method that transforms the challenge of identifying an optimal learning rate into selecting the most effective adaptive iterations for tuning GBDT. They established a theoretical connection between drift severity and the model's convergence rate. Based on this, they developed a new

Table 10
Provides a summary of current models designed for predicting data streams in the context of Intrusion Detection Systems (IDS), specifically focusing on recognizing and addressing Concept Drift (CD).

No	Article	Method/Classifier	Incremental Learning	Transfer learning	CD Detection Method	Variants of CD which evaluated					Dataset
						Sudden	Incremental	Gradual	Recurrent	Blip	
1	(Seth et al., 2024)	Adaptive Random Forest (ARF)			Hoeffding's bounds with moving average-test	√	×	√	×	×	CIC-IDS 2018
2	Andresini et al. (2021)	Semi-supervised intrusion detector	√	√	Incremental learning	×	×	×	×	×	CICIDS-2017
3	Uccello et al. (2024)	Unsupervised Concept Drift Detectors	×	×	Page-Hinkley Test, ADWIN	√	×	√	×	×	CIC-IDS2017, CSE-CIC-IDS2018
4	(Folino et al., 2020)Fino et al. (2020)	GP-based combiners	√	×	ADWIN, STEPD, DDM	×	×	×	×	×	ISCX IDS,
5	Shyaa et al. (2023)	OSELM, FA-OSELM and KP-OSELM	√	×	DDM	√	√	√	√	×	KDD Cup '99, CICIDS-2017, CSE-CIC-IDS-2018, and ISCX2012
6	Soltani et al. (2024)	CNN and LSTM	√	×	×	×	×	×	×	×	CICIDS-2017, CSE-CIC-IDS-2018
7	Rajeswari et al. (2022)	(ADWIN-SVM)	×	×	ADWIN	√	×	×	×	×	KDD Cup (1999). CICIDS 2017
8	Alqahtani (2024)	Adaptive Random Forest and Adaptive One-Class SVM	√	×	ADWIN	√	×	√	×	×	CICIDS 2017, DAPT 2020, InSDN, APT-SDN
9	(Alsuwat et al., 2023)	multilayer perceptron gradient decision tree (MLPGDT) classifiers.	×	×	(SAW_WDA)	×	×	×	×	×	DDoS 2016, UNSW-NB15, CICIDS 2017
10	(Abdel Wahab, 2022)	DNN	√	×	PCA	×	×	×	×	×	DS2OS traffic traces
11	Adnan et al. (2020)	Hyper heuristic	×	×	×	×	×	×	×	×	KDD Cup '99, NSL-KDD Landsat
12	(Yang et al., 2021)	Performance Weighted Probability Averaging Ensemble (PWPAE)	√	×	ADWIN, DDM	√	×	√	×	×	IoTID20 and CICIDS2017
13	(Seth et al., 2021)	Adaptive Random Forest classifier with ADWIN	√	×	ADWIN	×	×	×	×	×	CIC-IDS 2018
14	(Martindale et al., 2020)	heterogeneous ensemble classifiers	×	×	ADWIN	×	×	×	×	×	KDD Cup '99
15	(Mahmodi et al., 2020)	Fusion of online learning algorithms (Linear-order, Gaussian order)	×	×	Online fusion of experts	√	×	√	×	×	NSL-KDD, ISCX
16	(Qiao et al., 2022)	CNN and LSTM	×	×	Dynamic Residual Projection Method (DRPM)	×	×	×	×	×	Bot-IoT, UG-2C-5D
17	Yang and Shami (2023)	Ensemble classifiers	√	×	ARF-ADWIN, ARF-EDDM	√	×	√	×	×	CICIDS-2017, IoTID '20
18	dos Santos et al. (2022)	Reinforcement Learning	√	√	×	×	×	×	×	×	MAWIFlow
19	Jain and Kaur (2021)	Random Forest and Logistic Regression, Support Vector Machine, K-means clustering.	√	×	Sliding window	√	√	√	×	×	CIDDS-2017, NSL-KDD, Testbed
20	Kuppa and N.-A. Le-Khac (2022)	Nearest Class Mean (NCM)	√	×	Detection using autoencoder	×	×	×	×	×	Network and URL datasets
21	Xu et al. (2023)	Adaptive anomaly detection model	√	×	×	√	×	√	×	×	INSECTS, SWaT, WADI, BATADAL

(continued on next page)

Table 10 (*continued*)

No	Article	Method/Classifier	Incremental Learning	Transfer learning	CD Detection Method	Variants of CD which evaluated					Dataset
						Sudden	Incremental	Gradual	Recurrent	Blip	
		toward concept drift, ADTCD									
22	(Werner et al., 2021)	The Concept of Learning for Intrusion Event Aggregation in Real-time (CLEAR)	×	×	TSSD-EWMA	×	×	×	×	×	Collegiate Penetration Testing Competition (CPTC)
23	(Shao et al., 2021)	Ensemble classifier	√	×	ADWIN	×	×	×	×	×	Real traffic data

drift adaptation technique termed Adaptive Iterations (AdIter). This method autonomously determines the iteration count based on varying drift severities, enhancing prediction accuracy in data streams affected by concept drift. The study (Oliveira et al., 2023) comprehensively analyses handling both virtual and real concept drifts in data streams, focusing on their unique impacts on classifier suitability. To address these drifts effectively, the paper introduces the Online Gaussian Mixture Model with Noise Filter for Handling Virtual and Real Concept Drifts (OGMMF-VRD). This novel approach represents an advancement in managing the complexities associated with concept drift in data streams.

The study (Z. Li et al., 2020b) introduces the Dynamic Updated Ensemble (DUE), a novel algorithm for efficiently learning imbalanced, nonstationary data streams with concept drift. DUE stands out by learning in chunks without prior data reliance, effectively adapting to class shifts, and maintaining high efficiency. Its effectiveness is proven through tests on diverse datasets. The study (Q. Li et al., 2021a) introduces MIS-ELM, a new machine for classifying mixed data streams with limited labeled samples. It combines soft one-hot encoding and incremental learning, excelling in real-time processing.

The SCHCDOE algorithm, as discussed in the study (Liu and Zhao, 2023), represents a sophisticated approach to streaming data classification, adeptly addressing concept drift. It innovatively categorizes data states, employing tailored ensemble and incremental learning techniques, ensuring agile adaptation to dynamic data environments. Investigating the integration of ADODNN with ADWIN, The research (Priya and Uthra, 2023) delves into concept drift detection within streaming data, emphasizing enhancements in handling imbalanced and dynamic datasets. The study (Sun et al., 2021) introduces a novel Cost-Sensitive Data Stream (CSDS) classification method, utilizing unique cost-sensitive strategies and showcasing its adaptability across various streaming data scenarios. In (Talapula et al., 2023), the researchers developed an optimized hybrid deep learning classifier, combining deep Long Short-Term Memory (LSTM) and Recurrent Neural Networks (RNN) to detect concept drifts in streaming data efficiently. The research (Ancy and Paulraj, 2020) explores the HIDC approach, a hybrid solution for managing imbalances and concept drift in data streams, focusing on strategies like optimal reservoir allocation and resampling. In the study (Abbasi et al., 2021), ElStream emerges as a versatile tool for detecting concept drift in dynamic data streams, enhancing performance in various practical applications. Lastly, the article (Jiao et al., 2022) presents the DES-ICD approach for dynamic ensemble selection in imbalanced data streams, targeting efficient management of concept drift. Table 11 provides a consolidated overview of these techniques, categorizing them based on classifier type, application scope, methodology, drift detection strategy, and the specific CD variant evaluated.

3.3. *Dynamic feature selection*

Dynamic feature selection constitutes a vital approach in addressing the challenge of Virtual drift (Feature drift) within the context of concept drift detection and adaptation in machine learning. This phenomenon refers to the situation where the significance or utility of features in a dataset changes over time, consequently impacting the model's predictive efficacy. To tackle this issue, dynamic feature selection techniques are employed to dynamically identify and retain the most informative and discriminative features while diminishing the relevance of those that have become less salient. These methods typically entail continuously monitoring feature drift through statistical tests or heuristics and adjusting the feature set accordingly.

Dynamic feature selection empowers machine learning models to sustain their accuracy and generalization capabilities in evolving environments, enhancing their resilience and adaptability to shifting data distributions and feature dynamics. This exploration delves into recent dynamic feature selection methodologies advancements spanning feature importance measures, adaptive boosting, incremental learning, and fuzzy inference frameworks.

(Sahmoud and Topcuoglu, 2020) showcases a novel DFBFS algorithm adept at handling feature drifts in data streams, surpassing existing classification accuracy and drift recovery algorithms. The research presented in the study (Rabash et al., 2023) introduces the Enhanced Dynamic Filter-Based Feature Selection (E-DFBFS) framework for Intrusion Detection Systems (IDS), incorporating Multi-Objective Optimization via Metaheuristic Search (MOO-MHS). Utilizing the Non-Dominated Sorting Genetic Algorithm II (NSGA-II) for the periodic selection of solutions through both mean and median methods, E-DFBFS proposes an advanced strategy for dynamic feature selection, aiming to improve the effectiveness and precision of IDS.

(Noori et al., 2023) introduced a Dynamic Feature Selection (DFS) method using multi-objective PSO for intrusion detection. This approach addresses feature drift and high-dimensional data, enhancing efficiency and reducing overfitting. However, it requires substantial memory and involves complex implementation. Evaluations on HIKARI 2021 and TON_IoT 2020 datasets demonstrated its superior performance compared to existing methods.

(Wang, Wang and Wu, 2022) delves into dynamic feature weighting in data streams, harnessing a drift detection scheme based on the log-likelihood divergence score. The Adaptive QuickReduct algorithm (Ferone and Maratea, 2021) is proposed as an unsupervised algorithm to detect feature drifts. Its design prioritizes minimal memory usage and swift responsiveness. (Pishgoo et al., 2022) introduces the H-DIFS architecture for anomaly detection, amalgamating batch and stream processing strengths while employing a dynamic feature selection grounded in genetic algorithms.

(Barddal et al., 2019a) proposed a data stream dynamic feature selection technique named Adaptive Boosting for Feature Selection (ABFS). This method integrates chaining decision stumps with drift detectors within a boosting framework, catering to batch learning (Agrahari and Singh, 2022a). presents PCA-FDD, an adaptive method using principal component analysis to detect feature drift in predictive models. This approach identifies key feature subsets, monitors their changes over time, and helps forecast and correct prediction errors in the base model.

The article (Xu et al., 2022) introduces a sophisticated Online Group Streaming Feature Selection technique named FNE-OGSFS. This method

Table 11
Overview of existing models for stream data prediction with awareness and handling of CD with the variants of drifts used for evaluation.

No	Article	Method/Classifier	Incremental Learning	Application	Transfer learning	CD detection method	Variants of CD which evaluated					Dataset
							Sudden,	Incremental	Gradual	Recurrent	Blip	
1	Chen and Dai (2024)	Continuous Kernel Learning with Ensemble Methods	√	General	×	FHDDM	✓	✓	✓	×	×	SINE1,MIXED, CIRCLES, LED, INincreabrupt, INincreoccurring, Shuttle, CoverType, Poker, Electricity
2	Karimian and Beigy (2023)	Multi-source domain adaptation	×	General	×	CDDA	✓	✓	✓	×	×	SEA-a, SEA-g, LED-a, LED-g, Sine-a Sine-g, RBF-m, RBF-f, HYPER, PKRHND, ELEC.
3	Chen et al. (2024)	Dual-layer variable sliding window (DLVSW)	√	General	×	sliding window	✓	×	✓	×	×	BMS- Webview-1, T20I6D200K-X
4	(Mahdi et al., 2021h)	Ensemble classifier	√	General	×	Not specified	√	×	√	√	×	Wave, Sea, RBF, Tree, Electricity, Airline,
5	Suryawanshi et al. (2023)	Ensemble classifier	√	General	×	Page-Hinckley statistical test	✓	×	✓	×	×	Airline and SEA
6	Gâlmeanu and Andonie (2021)	SVM	√	General	×	ADWIN	√	×	√	×	×	SINE1, CIRCLES, COVERTYPE
7	Museba et al. (2021)	Ensemble classifier	√	General	×	EDDM	√	×	√	√	×	KDD 99, Sensor Data, Airlines, Poker Hand Cover type
8	Wu et al. (2022)	Random forest	×	General	×	Not specified	√	×	√	√	×	Electricity Rain, Airlines Covtype, Sensor
9	Wu et al. (2021)	Random forest	×	General	×	PROACTIVE DRIFT DETECTION	√	×	√	×	×	Agrawal, Random Tree, Covtype, Sensor
10	Chiu and Minku (2022)	Ensemble classifier	√	General	×	Clustering in the Model Space (CDCMS).	√	×	√	√	×	KDD cup99, Sensor, Power Supply, Sine1, Sine2, Agr1, Agr2, Agr3 Agr4, SEA1, SEA2, STA1, STA2,
11	Sarnovsky and Kolarik (2021)	Heterogeneous ensemble classifiers (Naive Bayes, k-NN, Decision trees)	√	General	×	Not specific	√	×	√	×	×	KDD 99, ELEC, Airlines, AIRL, COVT MIXEDBALANCED WAVEFORM, RBF
12	Cano and Krawczyk (2020)	Ensemble classifier	√	General	×	Kappa Updated Ensemble (KUE)	√	×	√	√	×	RBF, Random Tree, Agrawal, Asset Negotiation, LED, Hyperplane, Mixed, SEA, Sine, STAGGER, Waveform.
13	Bakhshi et al. (2023)	Broad Ensemble Learning System (BELS)	√	General	×	Major accuracy decline	√	√	√	×	×	Electricity, Mg2c2d, Usenet, Rotating Hyperplane, Spam, Email, Phishing, Poker, Usenet, Weather, Interchanging Rbf, Moving Squares.
14	Wang et al. 2022b	Elastic gradient boosting decision tree (eGBDT)	√	General	×	adaptive iterations (AdIter)	√	√	×	×	×	Electricity, Weather, Spam, Usenet1, Usenet2, Airline, SEAa, RTG, RBF, HYP, AGR
15	(Oliveira et al., 2023)	OGMMF-VRD	√	General	×	Gaussian Mixture Model	✓	✓	✓	×	×	Virtual 9, Circles, Sine2, PAKDD, NOAA
16	(Z. Li et al., 2020b)	Dynamic Updated Ensemble (DUE)	√	General	×	component weighting mechanism	✓	×	✓	✓	×	SEA, Hyper, RanRBF, Weat, Elec, Reve.
17	(Q. Li et al., 2021a)	(MIS-ELM)	√	General	×	(HDDDM)	×	×	×	×	×	Adult, Census, Credit-a, Credit-g, NSL-KDD2, NSL-KDD5, Mushroom, Nursery.
18	Liu and Zhao (2023)	Online ensemble classifier	√	General	×	sliding window	✓	✓	✓	×	×	SEAs, SEAG and SEAm, Hyperplane, Mixed paper electricity, Weather.
19	Priya and Uthra (2023)	Deep Neural Network	√	General	×	ADWIN	×	×	×	×	×	Chess, KDDcup99, Spam.

(*continued on next page*)

Table 11 (*continued*)

No	Article	Method/Classifier	Incremental Learning	Application	Transfer learning	CD detection method	Variants of CD which evaluated					Dataset
							Sudden,	Incremental	Gradual	Recurrent	Blip	
20	Sun et al. (2021)	Cost-Sensitive ReliefF algorithm (CS-ReliefF)	✓	General	×	DDM	✓	×	✓	×	×	HyperPlane, SEA, LED, Rotating Spiral, Spam, Sensor, Electricity, Airlines.
21	Talapula et al. (2023)	Hybrid deep learning classifier	✓	General	×	Optimized Key windowing (OKW)	×	×	×	×	×	Apache, Hadoop, Linux, Spark, Cloud Monitoring.
22	Ancy and Paulraj (2020)	Ensemble classifier	✓	General	×	Component weighting mechanism	✓	×	✓	×	×	Weather
23	Abbasi et al. (2021)	Ensemble classifier	✓	General	×	Majority Voting Technique	✓	×	✓	×	×	CoverType, Electricity, PokerHand, SEA, HYPERPLANE, RandomRBF, LED
24	Jiao et al. (2022)	Dynamic ensemble classifier	✓	General	×	DES-ICD	✓	✓	✓	×	×	SEA,SINe, TREE, Wave, Hype, Elec, GMSC, weather, Covtype

utilizes uncertainty measures based on fuzzy neighborhood entropy in conjunction with a Lasso model, facilitating efficient redundancy analysis. It is especially capable of managing dynamic feature selection challenges within data streams. The study (Chanu et al., 2023) introduced a hybrid feature selection strategy employing a voting mechanism. This approach integrates a Multilayer Perceptron with a Genetic Algorithm (MLP-GA). The primary advantages of this hybrid methodology include effectively reducing feature dimensions, eliminating redundant features, and identifying the most pertinent features.

Nancy (2020) integrated a dynamic feature selection model with a fuzzy inference system-based intelligent decision tree. This recursive algorithm collaboratively works with a support vector machine classifier to produce a ranked feature set.

This recursive algorithm collaboratively works with a support vector machine classifier to produce a ranked feature set (Wei et al., 2020). introduced a novel metric termed Dynamic Feature Importance (DFI), complemented by the Dynamic Feature Importance Feature Selection (DFIFS) algorithm. For enhanced accuracy using fewer features, the modified Dynamic Feature Importance-based Feature Selection (M-DFIFS) algorithm merges DFIFS with classical filters.

Lastly, the article (Ahsan et al., 2021) offers a dynamic feature selection algorithm utilizing supervised learning, notably XGBoost and random forest classifiers, to bolster network intrusion detection systems.

Table 12 explains the various approaches to dynamic feature selection.

3.4. Reinforcement learning for dynamic feature selection

Reinforcement learning (RL) has demonstrated considerable potential in addressing the challenge of dynamic feature selection. It enables the selection process to adapt dynamically to changes in data distribution. RL agents acquire the ability to make optimal decisions regarding feature selection through trial and error. This has the potential to enhance classification performance while reducing computational costs. The utility of reinforcement learning in dynamic feature selection lies in its capacity to adapt to shifting data distributions and optimize feature selection through iterative learning. Various RL-based approaches, including Q-learning and Deep Reinforcement Learning, have been introduced to tackle this issue. Traditional Q-learning methods have been widely employed in numerous studies.

Researchers have utilized deep Q-learning network (DQN) agents for Dynamic Feature Selection (DFS). A study (K. Liu et al., 2021a) introduced a framework comprising multiple agents responsible for a single feature. The combination of their decisions influences the selection of features. These agents define their state through descriptive statistics, and their reward is calculated as a weighted average, considering factors like relevance, redundancy, and precision. This approach trains DQNs through experience replay buffer sampling using Gaussian Mixture Models (GMM). Nonetheless, this method lacks an effective system for managing the trade-off between exploration and exploitation. Its efficiency declines as the feature count grows, which is attributed to the direct correlation between the number of features and the DQN size developed. Another research (Fan et al., 2020) presented an approach involving interactive reinforcement learning (IRL) and decision tree feedback (DTF) for navigation guidance. This method employs a Graph Convolutional Network (GCN) for state representation, integrating graph and tree structures. The agent's rewards are tailored based on the significance of features within the decision tree. In (Fang et al., 2019), a new architectural design was introduced for Deep Q-learning-based Feature Selection (DQFSA) targeting malware identification. Here, the Q-learning agent functions sequentially, with the precision of machine learning classifiers as its reward metric. Lastly (X. Li, 2021b), suggested a unique Deep Q-Network (DQN) model for feature selection. This model involves a solitary agent, represented by the DQN, selecting individual features sequentially. The reward mechanism is defined through a segmented equation comprising four distinct branches.

Table 12
The various approaches to dynamic feature selection.

No	Reference	Basic Techniques	Developed Technique	Advantages	Limitations	Application	Incremental Learning	Transfer learning	Dataset
1	Sahmoud and Topcuoglu (2020)	NSGA-II	DFBFS	Dealing with Pareto front for feature selection	Sensitivity to parameters	General	×	×	BGFD1-3, SEAFD
2	Rabash et al. (2023)	DFBFS	E-DFBFS	Selecting a solution automatically from the Pareto front	Sensitivity to population initialization	IDS	✓	×	KDD99, NSL-KDD
3	Noori et al. (2023)	Multi-objective Particle Swarm Optimization (PSO)	Dynamic Feature Selection (DFS) with Variable-Length PSO	Handles feature drift,	High resource use, Complex implementation	IDS	✓	×	HIKARI 2021, TON_IoT 2020
4	(Wang et al., 2022b)	Naive Bayes	DFWLR	Improved classifier accuracy	computational overhead	General	✓	×	Airlines, email_data, poker-ls, KDD99, Spam, GMSC. Nursery, EEG, COVTYPE,
5	Ferone and Maratea (2021)	QuickReduct	Adaptive QuickReduct	Detects a feature shift Successfully	Not specified	General	×	×	Electricity, GMSC
6	(Pishgoo et al., 2022)	(FSR), (BMB)	H-DIFS	effective in removing redundant and irrelevant features	A complex integration of batch and stream	General	✓	×	KDD Cup99, N-BaIoT, Forest Covertype, MiniBooNE
7	Barddal et al. (2019a)	Ada-Boost	ABFS	Identifies relevant features, improves classification rates	Adapting to feature drift dynamically	General	✓	×	COVTYPE, IADS, NOMAO, PAMAP2, Spam
8	Agrahari and Singh (2022a)	(PCA)	PCA-FDD	Detects feature drift by observing changes in percentage similarities.	Not specified	General	×	×	Digits08, Digits17, Wine, Forest Cover Type, Electricity, Phishing, Spam
9	Xu et al. (2022)	Fuzzy neighborhood rough sets model	(FNE-OGSFS)	Better performance in terms of stability and choosing the features.	Parameter tuning is needed.	General	×	×	SONAR, IONOSPHERE, COLON, LEUKEMIA, OVARIAN CANCER, MADELON, ARCENE.
10	Chanu et al. (2023)	MLP	(MLP-GA)	Reduces feature dimensions and removes redundancy.	Ignores initial feature dependencies and complexity.	General	×	×	CICDDoS 2017, NSL-KDD, DARPA
11	(Nancy, 2020)	(RFSA)	(DRFSA)	Enhanced intrusion detection accuracy	Not specified	IDS	×	×	KDD Cup, Network Trace
12	Wei et al. (2020)	Gini Importance, MIC	M-DFIFS	Achieves higher classification accuracy	Computational Time	General	×	×	RELATHE, lymphoma, ORL, Yale, GLI-85, WarpAR10P, Prostate-GE, ALLAML, CLL-SUB-111, TOX-171
13	Ahsan et al. (2021)	(ANOVA F-test), XGBoost	Dynamic Feature Selector	Reduce the number of features and improve classifier accuracy	Needs to adjust hyper-parameters of meta-learner.	IDS	×	×	NSL-KDD, UNSW-NB15

In the study (Ren et al., 2023), the Multi-Agent Feature Selection Intrusion Detection System (MAFSIDS) employs a dual-component architecture comprising a feature self-selection algorithm and a DRL-based attack detection module. The feature self-selection algorithm utilizes a multi-agent reinforcement learning framework to represent the traditional feature selection space, thus reducing model complexity. A Graph Convolutional Network (GCN) is also implemented for feature extraction. The DRL-based attack detection module incorporates the Mini-Batch technique for data encoding, enabling reinforcement learning in a supervised context.

In their study (Fan et al., 2021), the authors introduced a novel framework for Group-based Interactive Reinforced Feature Selection (GIRFS). This approach strikes a balance between single-agent and multi-agent Reinforced Feature Selection (RFS) strategies. Here, features are categorized into groups based on their similarities, with each group assigned its decision-making agents. Additionally, a hierarchical, teacher-like entity offers external action guidance to these agents. In another research (Paniri et al., 2021) , a feature selection technique was developed, leveraging Ant Colony Optimization (ACO) and complemented by the Temporal Difference (TD) reinforcement learning algorithm to enhance heuristic learning. Within this framework, features are equated to states (S), and the exploration of new features by each 'ant' constitutes a series of actions, with rewards determined based on two specific criteria following their actions.

Research (Xu et al., 2021) presented a unique approach involving creating a discriminant function using Q-learning. Here, the agent is responsible for selecting classification actions and refining the function based on feedback from its decisions. However, this method is confined to linear discriminant functions, which limits its effectiveness to more straightforward datasets. Advanced agents like the deep deterministic policy gradient (DDPG) have been developed to address more complex scenarios. In a study (Cheng et al., 2021), the feature selection process was conceptualized as a Markov Decision Process (MDP) and framed as a Reinforcement Learning (RL) challenge. They introduced a Deep Reinforcement Learning-based Feature Selector (DRLFS) and proposed a dynamic randomness policy to balance exploration and exploitation effectively. Additionally, they implemented two distinct search protocols: a fixed depth search and an adaptive depth search. Finally, in (Wu et al., 2023), a framework utilizing the Double DQN (DDQN) algorithm was presented for classifying Android malware. The decision-making network in DDQN incorporated Recurrent Neural Networks (RNN) to manage varying input lengths, and word embedding was used for feature representation, allowing for an assessment of semantic relevance.

The study (Kareem Thajeel et al., 2023) presents the Deep Q-Network Multi-Agent Feature Selection (DQN-MAFS) method for dynamic feature selection, addressing feature drift. This approach utilizes individual agents for each feature, making selection decisions, and incorporates the Fair Agent Reward Distribution-based Dynamic Feature Selection (FARD-DFS) to distribute rewards among agents fairly. These reviewed methods are summarized in Table 13, focusing on critical elements of RL, including state, action, reward, agent type, and the balance between exploration and exploitation. Notably, the existing methods still need to provide a comprehensive algorithm for fair reward distribution in multi-agent scenarios.

3.5. Real, virtual drift, data imbalance and high dimensionality in IDS

The domain of Intrusion Detection Systems (IDS) stands at the forefront of ensuring cybersecurity in the face of escalating and evolving threats. The efficacy of IDS hinges on their ability to recognize known attack patterns and adapt to new and emerging anomalies. This adaptability is continually challenged by phenomena such as concept drift. Concept drift bifurcates into real and virtual sub-types, each presenting distinct challenges to IDS methodologies. Real drift manifests when the underlying data distribution changes, necessitating an evolution in the detection model. On the other hand, virtual drift (Feature drift) occurs when the context of data or its interpretation varies over time without a

Table 13
Literature Survey of the existing algorithms in RL-based dynamic feature selection.

No	Article	Multi agent	Multi-level	State	Action	Reward	Reward distribution	Transfer learning	Agent	Exploration Exploitation Balance	Handling data unbalance	Classifier
1	(K. Liu, 2021a)	Yes	No	Statistical descriptors	Binary decision	Based on accuracy, redundancy, and relevance	No	No	DQN	$\varepsilon-$ greeduy	No	DT, SVM, XGBoost
2	Fan et al. (2020)	Yes	No	Graph Convolutional Network (GCN)	Binary decision	DT-based personal rewarding and accuracy from hybrid teaching	No	No	DQN	$\varepsilon-$ greeduy	No	Decision tree
3	Fang et al. (2019)	Yes	No	Selected feature	Select/ deselect	Accuracy	No	No	DQN	$\varepsilon-$ greeduy	No	Various classifiers
4	(X. Li et al., 2021b)	×	No	Binary state	Select or not select	Piecewise, based on the accuracy	No	No	DQN	$\varepsilon-$ greeduy	No	KNN
5	Ren et al. (2023)	Yes	No	Feature	Binary decision	Based on accuracy	Yes	No	DQL	Network policy	No	DQL
6	Fan et al. (2021)	Yes	No	Graph Convolutional Network (GCN)	Select or not select	Based on accuracy	No	No	Not Applicable	$\varepsilon-$ greeduy	No	Decision tree
7	(Paniri et al., 2021)	Yes	No	Feature	Select or not select	Cos similarity and correlation	No	No	Q-learning	Polynomial decay rate	No	KNN
8	Xu et al. (2021)	No	No	Feature	Class	Fixed reward/ punishment values	No	No	Q-learning	$\varepsilon-$ greeduy	No	Linear discriminant function
9	Cheng et al. (2021)	No	No	Encoding of the subset of selected features	Binary decision	Error decline	No	No	DDPG	Dynamic randomness policy	No	SVM
10	Wu et al. (2023)	No	No	Binary state	sequential selection by RNN	Accuracy	No	No	DDQN	$\varepsilon-$ greeduy	No	Various classifiers
11	Kareem Thajeel et al. (2023)	Yes	No	Binary state	Binary decision	(FARD-DFS)	Yes	No	DQN	Decay ε greedy	No	Binary

fundamental change in the underlying distribution. High dimensionality and data imbalance further exacerbate these challenges, presenting additional computational hurdles and the need for sophisticated feature selection mechanisms. Table 14 elucidates a spectrum of approaches, from semi-supervised models to complex ensemble classifiers and adaptive algorithms. Each entry in the compendium underscores the methodological innovations and the limitations that persist, such as the lack of update mechanisms, consideration of feature drift, or issues of computational efficiency and data imbalance. By examining these efforts, our survey critically addresses the contemporary state of IDS in combating both real and virtual drift while grappling with the intrinsic challenges posed by data imbalance and high-dimensional data streams.

4. Proposed framework

The framework depicted in Fig. 10 addresses three fundamental challenges in the realm of Intrusion Detection Systems (IDS): imbalance, high dimensionality, and drift, which are critical for maintaining the effectiveness and accuracy of IDS in dynamic environments.

1. **Imbalance Handling**: The first stage of the framework focuses on data balancing. This preprocessing step deals with the skewed distribution of class labels in the dataset. Imbalanced datasets can lead to biased models that perform poorly on minority classes, often the most critical to detect in the context of IDS. Techniques like oversampling, under-sampling, or synthetic data generation are employed to ensure that the training data is well-balanced and that all classes are adequately represented.
2. **High Dimensionality Handling:** The framework addresses high dimensionality through dynamic feature selection. High dimensionality can lead to the curse of dimensionality, where the feature space is so large that the available data becomes sparse. This sparsity makes it difficult for models to learn effectively and increases the risk of overfitting. Dynamic feature selection aims to identify and retain the most relevant features for the IDS task, reducing dimensionality and improving model performance.
3. **Classifier Update Mechanism:** The classifier update mechanism is pivotal to the framework. It allows for selecting and integrating various classifier types (Type 1, Type 2, …, Type N) based on their suitability to the current data characteristics. Drift detection can trigger the update, ensuring the model stays relevant and practical.
4. **Ensemble and optimizer:** The ensemble is responsible for aggregating the classifiers with the help of the optimizer to improve the aggregation model. The optimizer operates by calculating errors. The error is calculated based on the difference between the ground truth (GT) and the predicted values.
5. **Drift monitoring:** The third stage tackles the concept of drift. Drift in data streams refers to changes in the underlying distribution of the data over time, which can render the learned model obsolete. This stage involves continuous monitoring of the data stream to detect drift. If a drift is detected, the framework initiates a classifier update to adapt to the new data distribution. This ensures that the IDS remains accurate over time.
6. **Prediction:** Finally, the framework provides a prediction output. If no drift is detected, the current classifier makes a prediction. If drift is detected, the classifier is updated before making a prediction.

5. Discussion on the proposed framework

The framework depicted in Fig. 10 addresses three fundamental challenges in Intrusion Detection Systems (IDS): imbalance, high dimensionality, and drift, which are critical for maintaining the effectiveness and accuracy of IDS in dynamic environments. Compared to existing methods outlined in Table 14, our framework offers a comprehensive solution by addressing these challenges more effectively. While many existing methods, such as those by (Seth et al., 2024; Andresini et al., 2021; Soltani et al., 2024) do not specifically address data imbalance, our framework incorporates techniques like oversampling, undersampling, and synthetic data generation to ensure balanced training data. In terms of high dimensionality, methods like those by (Rabash et al., 2023; Noori et al., 2023; Wang, Wang and Wu, 2022; Sahmoud and Topcuoglu, 2020) focus on dynamic feature selection but often lack scalability. Our framework improves this by integrating scalable, dynamic feature selection processes, ensuring the effective handling of high-dimensional data. Furthermore, while models like by (Rajeswari et al., 2022; Xu et al., 2023) are effective in specific scenarios, they lack the flexibility to adapt dynamically to changing data characteristics. Our classifier update mechanism ensures continuous adaptation, enhancing model relevance over time. Additionally, ensemble methods, such as those by (Yang and Shami, 2023), rely on fixed aggregation techniques. In contrast, our framework's integration of an optimizer for dynamic error calculation and adjustment provides a more robust and adaptive approach. Finally, while many existing IDS methods, including those by (Qiao et al., 2022; Alsuwat et al., 2023), address real drift (concept drift), they often neglect virtual drift (feature drift). Our framework's continuous monitoring and adaptive updating capabilities ensure comprehensive handling of both types of drift, maintaining model effectiveness. These improvements make our framework a comprehensive solution for enhancing the accuracy, reliability, and adaptability of IDS in dynamic and evolving environments, providing robust defense mechanisms against sophisticated cyber threats.

While our proposed framework offers significant improvements over existing methods, it is not without limitations. Incorporating dynamic feature selection, continuous monitoring, and adaptive updates introduces additional computational overhead. This may require more powerful hardware and increase processing time, potentially affecting real-time detection capabilities in resource-constrained environments. The framework's comprehensive approach, which integrates multiple techniques for handling imbalance, high dimensionality, and drift, can lead to increased complexity in implementation. This complexity may require more extensive expertise and resources, posing challenges for deployment in smaller organizations with limited technical capabilities. Although the framework aims to improve scalability, the dynamic nature of the feature selection and continuous adaptation mechanisms may still encounter challenges when scaling to very large datasets or extremely high-dimensional data spaces. Further optimization may be necessary to ensure consistent performance at scale. The framework's effectiveness relies heavily on the accurate and timely detection of concept and feature drifts. Any inaccuracies or delays in drift detection can compromise the adaptive capabilities of the IDS, potentially leading to periods of reduced detection accuracy. While the framework has been designed to address a broad range of scenarios, its evaluation in this study has been limited to specific datasets. The generalizability of the results to other types of data and different cyber threat landscapes remains to be thoroughly tested. Although our framework includes all necessary components for drift-aware IDS, the quantitative experimental evaluation and comparative analysis with state-of-the-art methods are planned for future work. This future research will provide a more detailed assessment of the framework's performance and practical implications. In summary, while our proposed framework represents a significant step forward in designing adaptive IDS, these limitations highlight further research and development areas to ensure its robustness and applicability across diverse environments and threat scenarios.

6. Experimental design, existing datasets in IDS, and evaluation metrics

In the realm of Intrusion Detection Systems (IDS), designing robust experiments is crucial for evaluating the effectiveness of detection methodologies. This process typically involves several key steps. First,

Table 14
Overview of IDS methods from the perspective of concept drift sub-types, dimensionality and data imbalance.

No	Article	Method/Classifier	Real drift (CD)	Virtual drift	High Dimensionality	Data Imbalance	IDS/cybersecurity Security	Data Stream
1	(Seth et al., 2024)	Adaptive Random Forest (ARF)	√	X	X	X	√	√
2	Andresini et al. (2021)	Semi-supervised intrusion detector	√	X	X	X	√	√
3	Uccello et al. (2024)	Unsupervised Concept Drift Detectors	√	X	X	X	√	√
4	Folino et al. (2020)	GP-based combiners	√	X	X	X	√	√
5	Shyaa et al. (2023)	OSELM, FA-OSELM and KP-OSELM	√	X	X	√	√	√
6	Soltani et al. (2024)	CNN and LSTM	√	X	√	X	√	√
7	Rajeswari et al. (2022)	(ADWIN-SVM)	√	X	X	X	√	√
8	Alqahtani (2024)	Adaptive Random Forest and Adaptive One-Class SVM	√	X	X	X	√	√
9	(Alsuwat et al., 2023))	Multilayer perceptron gradient decision tree (MLPGDT) classifiers	√	X	X	X	√	√
10	(Abdel Wahab, 2022)	DNN	√	X	√	X	√	√
11	Adnan et al. (2020)	Hyper heuristic	√	X	X	X	√	√
12	(Yang et al., 2021)	Performance Weighted Probability Averaging Ensemble (PWPAE)	√	X	X	X	√	√
13	(Seth et al., 2021)	Adaptive Random Forest classifier with ADWIN	√	X	X	X	√	√
14	(Martindale et al., 2020)	Heterogeneous ensemble classifiers	√	X	X	X	√	√
15	(Mahmodi et al., 2020)	Fusion of online learning algorithm	√	X	X	X	√	√
16	(Qiao et al., 2022)	CNN and LSTM	√	X	X	X	√	√
17	Yang and Shami (2023)	Ensemble classifiers	√	√	X	X	√	√
18	dos Santos et al. (2022)	Reinforcement Learning	√	X	X	X	√	√
19	Jain and Kaur (2021)	Random Forest and Logistic Regression, Support Vector Machine, K-means clustering.	√	X	X	X	√	√
20	Kuppa and N.-A. Le-Khac (2022)	Nearest Class Mean (NCM)	√	X	X	X	√	√
21	Xu et al. (2023)	Adaptive anomaly detection model toward concept drift, ADTCD	√	X	X	X	√	√
22	(Werner et al., 2021)	The Concept of Learning for Intrusion Event Aggregation in Real-time (CLEAR)	√	X	X	X	√	√
23	(Shao et al., 2021)	Ensemble classifier	√	X	X	X	√	√
24	Chen and Dai (2024)	Continuous Kernel Learning with Ensemble Methods	√	X	X	X	X	√
25	Karimian and Beigy (2023)	Multi-source domain adaptation	√	X	X	X	X	√
26	Chen et al. (2024)	Dual-layer variable sliding window (DLVSW)	√	X	X	X	X	√
27	(Mahdi et al., 2021)	Ensemble classifier	√	X	X	X	X	√
28	Suryawanshi et al. (2023)	Ensemble classifier	√	X	X	X	X	√
29	Gâlmeanu and Andonie (2021)	SVM	√	X	X	X	X	√
30	Museba et al. (2021)	Ensemble classifier	√	X	X	X	X	√
31	Wu et al. (2022)	Random forest	√	X	X	X	X	√
32	Wu et al. (2021)	Random forest	√	X	X	X	X	√
33	Chiu and Minku (2022)	Ensemble classifier	√	X	X	X	X	√
34	Sarnovsky and Kolarik (2021)	Heterogeneous ensemble classifiers (Naive Bayes, k-NN, Decision trees)	√	X	X	X	X	√
35	Cano and Krawczyk (2020)	Ensemble classifier	√	X	X	√	X	√
36	Bakhshi et al. (2023)	Broad Ensemble Learning System (BELS)	√	X	X	√	X	√

(continued on next page)

Table 14 (*continued*)

No	Article	Method/Classifier	Real drift (CD)	Virtual drift	High Dimensionality	Data Imbalance	IDS/cybersecurity Security	Data Stream
37	Wang et al. 2022b	Elastic gradient boosting decision tree (eGBDT)	√	X	√	X	X	√
38	(Oliveira et al., 2023)	Online Gaussian Mixture Model with Noise Filter (OGMMF-VRD)	√	√	X	X	X	√
39	(Z. Li et al., 2020b)	Dynamic Updated Ensemble (DUE)	√	X	X	√	X	√
40	(Q. Li et al., 2021a)	(MIS-ELM).	√	X	X	X	X	√
41	Liu and Zhao (2023)	Online ensemble classifier	√	X	√	X	X	√
42	Priya and Uthra (2023)	Deep Neural Network	√	X	X	√	X	√
43	Sun et al. (2021)	Cost-Sensitive ReliefF algorithm (CS-ReliefF)	√	X	X	√	X	√
44	Talapula et al. (2023)	Hybrid deep learning classifier	√	X	X	X	X	√
45	Ancy and Paulraj (2020)	Ensemble classifier	√	X	X	√	X	√
46	Abbasi et al. (2021)	Ensemble classifier	√	X	X	X	X	√
47	Jiao et al. (2022)	Dynamic ensemble classifier	√	X	X	√	X	√
48	Sahmoud and Topcuoglu (2020)	Dynamic Filter-Based Feature Selection (DFBFS)	X	√	√	X	X	√
49	Rabash et al. (2023)	Enhanced Dynamic Filter-Based Feature Selection (E-DFBFS)	X	√	√	X	X	√
50	Noori et al. (2023)	Dynamic Feature Selection (DFS) with Variable-Length PSO	X	√	√	√	X	√
51	(Wang et al., 2022b)	Dynamic feature weighting algorithm	X	√	√	X	√	√
52	Ferone and Maratea (2021)	Quick Reduct algorithm	X	√	√	X	X	√
53	(Pishgoo et al., 2022)	Dynamic and intelligent feature selection	X	√	√	X	X	√
54	Barddal et al. (2019a)	Adaptive Boosting for Feature Selection (ABFS)	X	√	√	X	X	√
55	Agrahari and Singh (2022a)	Principal component analysis-based feature drift detection (PCA-FDD)	X	√	√	X	X	√
56	Xu et al. (2022)	Fuzzy neighborhood entropy based on online group streaming feature selection (FNE-OGSFS)	X	√	X	X	X	√
57	Chanu et al. (2023)	Multilayer perceptron with genetic algorithm (MLP-GA)	X	√	X	X	X	√
58	Nancy (2020)	Dynamic recursive feature selection algorithm (DRFSA)	X	√	X	X	X	√
59	Wei et al. (2020)	Modified-Dynamic Feature Importance based Feature Selection (M-DFIFS)	X	√	X	X	X	√
60	Ahsan et al. (2021)	Dynamic Feature Selector	X	√	X	X	X	X
61	(K. Liu, 2021a)	Multi-agent reinforcement learning for the feature selection	X	X	√	X	X	X
62	Fan et al. (2020)	Interactive reinforcement learning (IRL)	X	X	√	X	X	X
63	Fang et al. (2019)	Deep Q-learning based Feature Selection Architecture (DQFSA)	X	X	√	X	X	X
64	(X. Li et al., 2021b)	Deep Q-Network	X	X	√	X	X	X
65	Ren et al. (2023)	Multi-agent feature Selection Intrusion Detection System (MAFSIDS)	X	X	√	X	√	X
66	Fan et al. (2021)	Group-based interactive Reinforced Feature Selection (GIRFS)	X	X	√	X	X	X
67	Paniri et al. (2021)	(ACO) with reinforcement learning	X	X	√	X	X	X
68	Xu et al. (2021)	Dynamic feature selection based on Q-Learning mechanism (QLFS)	X	X	√	X	X	X
69	Cheng et al. (2021)	Deep Reinforcement Learning based Feature Selector (DRLFS)	X	X	√	X	X	X
70	Wu et al. (2023)	(DDQN) algorithm	X	X	√	X	√	X
71	Kareem Thajeel et al. (2023)	Deep Q-network multi-agent (DQN-MAFS)	X	√	√	X	√	√

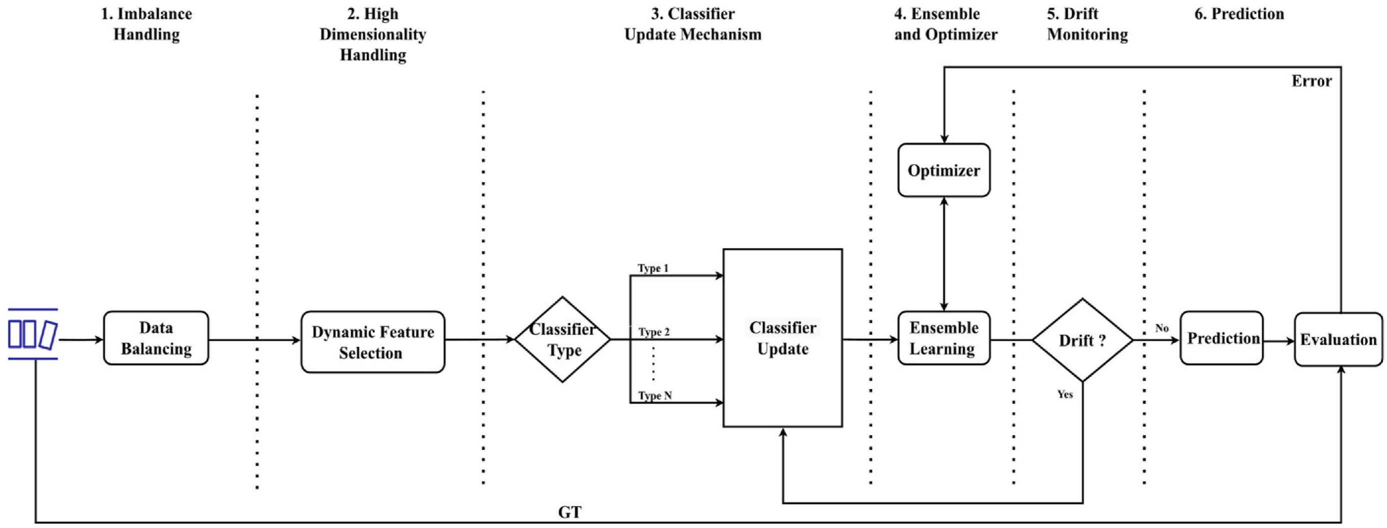

Fig. 10. Our proposed framework for drift-aware IDS.

the problem and objectives of the study are clearly defined, identifying the types of intrusions to detect and the aspects of IDS performance to evaluate. Next, appropriate datasets that reflect real-world network traffic and include a wide range of intrusion types are selected. Preprocessing these datasets ensures data quality and relevance, involving tasks like data cleaning, normalization, feature extraction, and feature selection. It is also essential to handle missing values and balance the dataset.

The selected machine learning models or IDS methodologies are then trained on the preprocessed dataset, typically by splitting the data into training and testing sets to avoid bias. To ensure robustness, the trained models are evaluated using appropriate metrics and validation techniques, such as cross-validation. Finally, the performance of different models or methodologies is compared, and the results are analyzed to identify strengths, weaknesses, and areas for improvement. Throughout this process, documenting the experimental setup, methodologies, results, and analysis comprehensively ensures transparency and reproducibility.

Numerous publicly available datasets are commonly used for evaluating IDS. The KDD Cup 1999 dataset is one of the earliest and most widely used, although it is now considered outdated (Stolfo et al., 2000). The NSL-KDD dataset is an improved version that addresses some of the original issues (Tavallaee et al., 2009). The UNSW-NB15 dataset, generated using the IXIA PerfectStorm tool, reflects contemporary network traffic and attack patterns (Moustafa and Slay, 2015). The CICIDS2017 dataset, created by the Canadian Institute for Cybersecurity, includes benign and malicious traffic reflective of real-world scenarios (Sharafaldin et al., 2018), and its updated version, CSE-CIC-IDS2018, expands on these features. For loT environments, the TON_IoT dataset includes various attack types on loT devices (Alsaedi et al., 2020). Evaluating IDS performance requires a comprehensive set of metrics. Accuracy measures the overall correctness of predictions but can be misleading in imbalanced datasets. Precision indicates the accuracy of positive predictions and is crucial when the cost of false positives is high. At the same time, recall measures the ability to detect actual positive instances, which is essential in minimizing false negatives. The F1-Score balances the trade-off between precision and recall. False Positive Rate (FPR) measures the rate of incorrect positive predictions, and False Negative Rate (FNR) measures the rate of missed detections. The Area Under the ROC Curve (AUC-ROC) provides an aggregate performance measure across all classification thresholds (Fawcett, 2006). The confusion matrix offers a detailed breakdown of classification results, facilitating a deeper understanding of model performance. By following these steps and utilizing these metrics, researchers can comprehensively assess IDS models, ensuring their effectiveness and reliability in detecting and mitigating cyber threats.

7. Recommendations for threat detection systems

Improving the reliability of threat detection systems is crucial for enhancing cybersecurity defences against increasingly sophisticated and evolving threats. This can be achieved through several strategic approaches supported by various research studies.

First, stream processing frameworks like Apache Kafka and Apache Flink can enhance real-time data processing for real-time data ingestion and processing. This allows threat detection systems to analyze data as it arrives, providing timely responses to emerging threats (Andresini et al., 2021). Implementing low-latency data analytics solutions ensures quick detection and response, with techniques such as in-memory computing employed to reduce processing delays.

Next, adaptive learning algorithms play a vital role. Incorporating incremental learning algorithms that update models with new data without retraining from scratch helps dynamically adapt to new threat patterns (Shyaa et al., 2023). Using online learning models allows continuous learning from new data streams, enabling the system to evolve with changing threat landscapes (Seth et al., 2024). Continuous model training can be achieved by using recent data to set up automated pipelines for regular model retraining, ensuring that detection models remain current and effective against new threats (Yang et al., 2021). Combining supervised, unsupervised, and reinforcement learning techniques creates more robust models capable of handling various data and threat scenarios (dos Santos et al., 2022).

Integrating threat intelligence enriches the detection system with the latest known threats and vulnerabilities information. Real-time threat intelligence feeds from reputable sources and participation in industry threat intelligence sharing platforms provide insights into emerging threats and collaborative defence strategies (Adnan et al., 2020).

Feature engineering and selection are critical for improving model accuracy and reducing computational overhead. Implementing dynamic feature selection techniques identifies and prioritizes the most relevant features for threat detection (Wang, Wang and Wu, 2022). Developing methods to detect and adapt to feature drift ensures that the features used by detection models remain relevant over time (Noori et al., 2023).

Ensemble learning techniques enhance detection accuracy and reduce false positives and negatives. Using ensemble methods such as bagging, boosting, and stacking combines multiple models' predictions (Cano and Krawczyk, 2020). Ensuring diversity in the ensemble models by combining different algorithms captures various aspects of threat behaviors (Bakhshi et al., 2023).

To improve robustness against evasion techniques, adversarial training involves using adversarial examples to enhance models' robustness against techniques attackers use to bypass security measures (Jiao et al., 2022). Incorporating behavioral analysis techniques detects anomalies in user and system behaviors, indicating sophisticated attacks that evade traditional signature-based detection (Mahdi et al., 2021).

Scalability and performance optimization are essential for handling large volumes of data and maintaining system efficiency. Designing detection systems using scalable architectures, such as cloud-based solutions and microservices, ensures scalability (Liu and Zhao, 2023). Continuous performance monitoring helps identify and address bottlenecks, maintaining high reliability and efficiency (Karimian and Beigy, 2023).

Advanced threat-hunting capabilities involve proactive techniques to search for indicators of compromise and potential threats within the network rather than relying solely on automated detection (Gâlmeanu and Andonie, 2021). Utilizing User and Entity Behavior Analytics (UEBA) identifies deviations from normal behavior patterns, revealing insider threats and advanced persistent threats (Shao et al., 2021).

Comprehensive logging and auditing ensure detailed logging of all network and system activities, providing a rich data source for threat detection and forensic analysis (Nancy, 2020). Regular audits of the detection system's performance help update detection rules and models based on audit findings, improving reliability (Wei et al., 2020).

Finally, addressing concept drift and feature drift awareness and handling involves implementing continuous monitoring techniques to detect concept drift, where the statistical properties of the target variable change over time (Xu et al., 2022). Techniques such as statistical process control charts or drift detection methods (e.g., DDM, EDDM) are used (Priya and Uthra, 2023). Developing and deploying algorithms to detect and adapt to feature drift, where the relevance of input features changes, involves periodic re-evaluation of feature importance and automated feature selection to ensure the most relevant features are used (Priya and Uthra, 2023). Adaptive learning algorithms adjust to concept and feature drift, with ensemble learning methods weighing models differently over time based on their performance (Ren et al., 2023). Establishing feedback loops continuously updates the models based on new data and detected drifts, ensuring the IDS remains effective even as the threat landscape evolves (Cheng et al., 2021).

By implementing these strategies, organizations can significantly enhance the reliability of their threat detection systems, ensuring they are better equipped to handle the dynamic and evolving nature of cyber threats.

8. Research challenges

The domain of artificial intelligence, mainly applied to cybersecurity, must continually evolve to counteract the adaptive and sophisticated threats posed by adversaries. IDS is at the epicentre of this battle, relying on data-driven models to detect and prevent unauthorized access or abuse of network resources. As network interaction complexity escalates, so does the complexity of maintaining the accuracy and relevance of these detection systems. This section confronts the significant research challenges that are formidable barriers to the next generation of IDS. Sub-section A addresses the first of these challenges. Concept drift and feature drift often occur simultaneously, each exerting a transformative impact on the performance of IDS. While concept drift denotes changes in the underlying patterns of malicious behavior, feature drift pertains to the change in the relevance of features over time. The joint occurrence can compound the difficulty of maintaining an effective detection model, yet the current literature shows a disconcerting gap in models that address both drifts in unison. This section underscores the imperative need for integrated approaches that can simultaneously accommodate the dynamism of both concept and feature drift. In Sub-section B, the focus shifts to the intersection of concept drift and another pervasive challenge: class imbalance. Many IDS face skewed distributions of attack classes, which can lead to biases in model training and, consequently, reduced detection performance, especially for minority classes that often represent the most dangerous attacks. The sub-section delineates the shortfall in current methodologies to jointly consider concept drift and imbalance, advocating for research endeavours that amalgamate solutions to these intertwined issues.

Lastly, Sub-section C highlights a gap in detecting complex drift types. Drifts can manifest in various forms, including incremental, abrupt, or gradual, each requiring specific strategies for detection and adaptation. The sub-section critiques the limited exploration into complex and compounded drift scenarios, which are becoming increasingly common in real-world data streams, urging a comprehensive investigation into robust detection mechanisms that can grapple with this complexity.

Through a focused discourse on these challenges, Section V aims to elucidate the current limitations within the field and foster a dialogue towards innovative solutions that can enhance the resilience and adaptability of IDS in an ever-changing threat landscape.

8.1. Ignorance of joint handling of concept drift and feature drift

Data stream classification has become a crucial learning issue in recent years, as more and more data can be treated as streaming data. Most of these data are characterized as high speed, nonstationary data distribution, and infinite length. The underlying data distribution of data streams constantly changes over time (Soltani et al., 2024). By far, the most widely researched task is Data stream classification or online classification; novel learning proposals must tackle the ephemeral property of streams, namely concept drift (Andresini et al., 2021; Folino et al., 2020), data streams are susceptible to different types of drifts: (i) changes in the characteristic feature values, (ii) evolution of the value domains of features over time, or (iii) features that were once relevant and that now may become meaningless or the other way around (Barddal et al., 2017; Agrahari and Singh, 2022a).

There is one specific type of drift that has not yet been thoroughly studied, namely feature drift (Barddal et al., 2019a), and we can consider it to be another challenging trait of data streams. Feature drift occurs whenever a subset of features becomes or ceases to be relevant to the learning task; thus, learners must detect and adapt to these changes accordingly (Rabash et al., 2023).

Reading the literature on dynamic feature selection to counter the impact of feature drift, we find that few approaches have been proposed in this context. However, they are limited in various aspects: in the work of (Sahmoud and Topcuoglu, 2020), there is a concern about the computational efficiency of the approach, which makes it not scalable for high dimension data which is also the same issue that is found as limitation of the work (Wang, Wang and Wu, 2022). Another matter is that the feature interaction and their exploration should have been addressed in most approaches (Barddal et al., 2019b).

Dealing with high dimensionality requires a dynamic selection of essential features instead of a fixed one, which could be more effective when concept or feature drift happens. Hence, it is necessary to reformulate the problem from a simple classification problem to learning and prediction with an awareness of evolving the records or chunks in the stream data and using an effective way to handle the issues of feature drift in the high dimensional space. Observing Table 14 we conclude that the literature lacks a method or algorithm that includes supporting batch learning, handling of feature and concept drift, and is applied and validated in IDS.

Out of 71 articles in Table 14, we found relatively few studies that have addressed feature drift (Rabash et al., 2023; Noori et al., 2023; Kareem Thajeel et al., 2023). The mentioned articles have not considered the issue of concept drift, which affects feature drift. The majority of the existing approaches focus on one of the two issues, concept drift or feature drift, ignoring the inter-dependency aspect and overlapping between them, and do not consider those two issues when designing a drift detection strategy. Hence, fulfilling the research gap is the final aim of this research.

To bridge this gap, future research should focus on developing integrated methodologies that simultaneously accommodate the dynamism of concept and feature drift. This includes creating algorithms that support batch learning, handle feature and concept drift, and are validated in IDS applications.

8.2. Lacking joint handling of concept drift and imbalance

A notable research gap in data stream mining is the need for comprehensive methodologies that concurrently address the dual challenges of concept drift and class imbalance. Concept drift studies often focus on models' adaptability to changing data distributions. At the same time, those addressing class imbalance are chiefly concerned with ensuring that minority classes are adequately represented in the model. Each issue has its own unique set of complexities, and yet they are frequently intertwined in real-world data streams. However, the prevailing literature tends to compartmentalize these challenges, offering solutions tailor-made for either concept drift or class imbalance. However, relatively few studies have addressed both (Jiao et al., 2022; Priya and Uthra, 2023). This siloed approach fails to deliver robust models capable of performing well in dynamic, imbalanced environments. The lack of integrated methodologies limits the theoretical development in this domain and impacts the practical applicability of data stream mining algorithms in complex, evolving settings.

Future research should aim to create hybrid models that simultaneously address concept drift and class imbalance. Such models would improve the accuracy and reliability of IDS, especially in detecting rare and critical attack patterns that are often underrepresented in training data. Techniques such as adaptive resampling, cost-sensitive learning, and ensemble methods can be explored to balance the class distribution while continuously adapting to concept drift. Additionally, developing synthetic data generation methods to simulate minority class samples dynamically could provide a more balanced training dataset, improving the robustness of IDS.

8.3. Lacking study of the detection of complex types of drifts

A significant research gap in data stream mining and concept drift detection pertains to the limited investigation into complex types of drifts that manifest as amalgamations of simpler drift forms. Existing research has predominantly focused on detecting and adapting to elementary forms of drift in data distribution, such as sudden and gradual Drifts (Seth et al., 2024; Uccello et al., 2024; Alqahtani, 2024; Xu et al., 2023; Chen et al., 2024), or sudden and incremental Drifts (Wang, 2022a). While these foundational studies have been indispensable in advancing our understanding of concept drift, they do not fully encapsulate the complexity of drift phenomena observed in real-world scenarios, where multiple types of drift often coalesce. For instance, a sudden shift in data distribution may be immediately followed by a gradual change or two or more types of drift could co-occur (Jain and Kaur, 2021; Karimian and Beigy, 2023). Current algorithms and detection mechanisms are generally ill-equipped to identify and adapt to multifaceted drift patterns. The dearth of studies in detecting complex composite drift types hampers theoretical growth. It poses a significant impediment to successfully deploying data stream mining models in environments characterized by dynamic and multi-layered changes. Therefore, there is an exigent need for future research to explore and formalize methodologies capable of identifying and handling complex combinations of drift types, effectively closing this research gap.

These methodologies should focus on creating more sophisticated drift detection algorithms that can accurately discern and react to subtle and multifaceted changes in data streams. Researchers should aim to develop hybrid models that integrate various machine learning techniques, such as reinforcement learning, deep learning, and ensemble methods, to enhance the robustness and adaptability of IDS.

9. Conclusion

This article explored the intricate challenges that delineate the current state of Intrusion Detection Systems (IDS) within the dynamic realm of cybersecurity. As the technological landscape advances, so does the sophistication of threats, necessitating a perpetual evolution of IDS to counteract these threats effectively. We delved into the complex interplay of concept drift and feature drift, the nuances of class imbalance, and the underexplored territory of complex drift types, thereby uncovering the formidable barriers that must be surmounted to pave the way for the next generation of IDS.

Our critical literature examination highlighted a significant research gap: the lack of models that concurrently addressed the dual phenomena of concept drift and feature drift. These drifts do not occur in isolation; their joint impact drastically diminishes the performance of IDS. An integrated approach that can navigate the intricacies of concept and feature drift is essential for developing sophisticated IDS. The intersection of concept drift with class imbalance presents an additional layer of complexity. Current methodologies often address these challenges in isolation, overlooking the compounded difficulty presented when they occur simultaneously. This oversight underscores a critical need for methodologies that encapsulate both phenomena to enhance the adaptability and accuracy of IDS in imbalanced data environments. There is a lack of research on detecting complex types of drifts, which is a significant issue. Real-world data streams often display multifaceted drift patterns that current detection mechanisms have difficulty identifying and adapting to. This necessitates a dedicated research effort to create robust algorithms that recognize and respond to complex drift scenarios.

We proposed a comprehensive framework that addresses the challenges of data imbalance, high dimensionality, and drift. This framework integrates data balancing techniques, dynamic feature selection, flexible classifier updates, ensemble learning, and drift monitoring to ensure robust and adaptive IDS operations. By incorporating these elements, these improvements underscore the necessity for developing adaptive and resilient IDS. Our study identifies under-represented areas in current research and sets a clear direction for future investigations. Future research could significantly advance the efficacy of IDS by addressing the joint handling of concept and feature drift, the interplay between concept drift and class imbalance, and the detection of complex drift types.

The domains of concept drift, feature drift, and class imbalance, particularly in the context of IDS, demand a more integrated and sophisticated approach to research. As the battle against cyber threats intensifies, it is incumbent upon the research community to forge innovative solutions that address these challenges individually and in their joint manifestation. Through such focused and collaborative efforts, we can enhance the resilience and adaptability of IDS, ensuring their effectiveness in the face of an ever-changing cyber threat landscape. By summarizing our core findings, explicitly stating our contributions, and emphasizing the significance and implications of our research, this conclusion aims to provide a clear and impactful summary for readers, underscoring the importance of advancing IDS technology to meet evolving cybersecurity challenges.

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CRediT authorship contribution statement

Methaq A. Shyaa: Writing – review & editing, Writing – original draft, Validation, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Noor Farizah Ibrahim:** Formal analysis, Investigation, Supervision, Validation, Writing – review & editing. **Zurinahni Zainol:** Writing – review & editing, Writing – original draft, Validation, Supervision, Project administration, Methodology, Formal analysis, Conceptualization. **Rosni Abdullah:** Writing – review & editing, Validation, Supervision, Formal analysis. **Mohammed Anbar:** Writing – review & editing, Supervision, Methodology, Formal analysis. **Laith Alzubaidi:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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